Online supplement to "Neural means and kernel corrections for operator learning"

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Abstract

This supplement accompanies the main text and is archived with the code repository. Section S1 states the finite-sample bounds behind every fitted stage of the pipeline; Section S2 the margin law and the per-coordinate selection results; Section S3 the effective-dimension identity and its Marčenko–Pastur closed form; Section S4 the additional structural-mechanics results (ablation, error localization, spectra, uncertainty quantification, cost); Section S5 the OCO-2 input-metric study and the data-scaling and ClimSim results; Sections S6–S8 all proofs, the implementation and compute record, and the numerical verification of every stated result. Numbering of the form Sn refers to this document; all other references point to the main text.

S1 Guarantees for the fitted stages

S1.1 Symmetry averaging does not increase the risk

The elasticity problem behind the benchmark is invariant under the reflection $x_1 \mapsto 1 - x_1$: the domain, the boundary partition, and the constitutive model are all symmetric, the input distribution has a reflection-invariant covariance, and reflecting the load reflects the stress field. On the discrete grid this is expressed by two permutation matrices: $S \in \mathbb{R}^{p\times p}$ reverses the load samples and $T \in \mathbb{R}^{q\times q}$ reverses the rows of the stress field. Both are orthogonal involutions. Section 2 describes a direct data-driven check of the equivariance $G(Su) = T\,G(u)$; we treat it as an assumption here.

Proposition S1.1 (symmetrization)**.** *Assume $G(Su) = TG(u)$ for μ-a.e. u, that μ is invariant under S, and that T is an orthogonal involution ($T^2 = I$). For a measurable $\widehat{G} : \mathcal{U} \to \mathcal{V}$ define its symmetrization*

$$\widehat{G}_S(u) \;=\; \tfrac{1}{2}\Big(\widehat{G}(u) + T\,\widehat{G}(Su)\Big).$$

Then, for every loss $\ell(\cdot, v)$ that is convex in its first argument and satisfies $\ell(Tw, Tv) = \ell(w, v)$,

$$\mathbb{E}_{u\sim\mu}\,\ell\big(\widehat{G}_S(u), G(u)\big) \;\le\; \mathbb{E}_{u\sim\mu}\,\ell\big(\widehat{G}(u), G(u)\big).$$

The relative error satisfies both hypotheses (T orthogonal gives $\ell(Tw, Tv) = \ell(w, v)$). Proposition S1.1 justifies the two uses of the symmetry in the pipeline: averaging each trained model with its reflected evaluation at test time can only help on average, and training on reflection-augmented data is ordinary empirical risk minimization for the symmetrized class. The measured effect is small and, being an expectation, need not hold sample by sample: over the thirty reflected members of the seed campaign it helps twenty-nine times and hurts once, by a thousandth of a point (Section S10). The convexity argument behind the proposition is the standard one for group-averaged predictors; Lyle et al. (2020) give a general account.

S1.2 What the kernel-flow regularizer estimates

The kernel-flow (KF) loss of Owhadi and Yoo (Owhadi and Yoo, 2019), in the ℓ^2 variant used by Yoo and Owhadi (2021) to regularize deep networks, enters our ablation study as an optional term in the training of the neural means. Its motivation is often stated informally ("a kernel is good if half the data predicts the rest"). The following observation makes the statement exact. Let $z_i = \phi_\theta(u_i)$ be the features of a batch $B = \{1, \ldots, b\}$ (b even), let k be a positive definite kernel on feature space, and for $C \subset B$, $|C| = b/2$, let I_C denote the k-interpolant of $\{(z_j, v_j) : j \in C\}$. The batch KF loss is

$$e_2(\theta; C) \;=\; \sum_{i \in B} \big\| v_i - I_C(z_i) \big\|_2^2. \tag{S1.1}$$

Proposition S1.2 (KF loss is a leave-half-out estimate)**.** *Let C be drawn uniformly among the subsets of B of size $b/2$ and assume k restricted to $\{z_i\}_{i \in B}$ is strictly positive definite. Then $e_2(\theta; C) = \sum_{i \in B \setminus C} \|v_i - I_C(z_i)\|_2^2$, and*

$$\mathbb{E}_C\big[e_2(\theta; C)\big] \;=\; \frac{b}{2}\, \mathbb{E}_{C,\, i \sim \mathrm{Unif}(B \setminus C)}\big[\big\|v_i - I_C(z_i)\big\|_2^2\big],$$

i.e. $\frac{2}{b}\, e_2$ is an unbiased estimator, over the split randomness, of the expected squared error committed on a held-out point by the kernel predictor trained on a random half of the batch. When (B, C) are resampled at every step, the stochastic gradient $\nabla_\theta e_2(\theta; C)$ is unbiased for $\nabla_\theta\, \mathbb{E}_{B,C}[e_2(\theta; C)]$ whenever the expectation and derivative commute (e.g. under local Lipschitz domination).

The training loss we minimize for a neural mean is a sum of an empirical risk term and, optionally, $\beta\, e_2$: the first term is a linear functional of the empirical measure of the batch and measures fit, while Proposition S1.2 shows the second measures the *generalization of a kernel method built on the learned features.* This is the sense in which a KF-regularized network is trained to be a good feature map for the kernel correction that follows.

S1.3 Stacking on a held-out split is safe

Between the means and the correction sits one more fitted object: the convex weights of the stacked ensemble, chosen to minimize the empirical risk on the validation split. Two elementary facts justify this step. First, by convexity, a convex combination of predictors is never worse than the corresponding weighted average of their risks, so stacking cannot be hurt by a weak member beyond its weight. Second, the weights live on a low-dimensional simplex and are fitted on hundreds of samples, so the selection cannot meaningfully overfit; the following proposition quantifies this with explicit constants.

Proposition S1.3 (validation stacking). *Let $f_1, \ldots, f_M : \mathcal{U} \to \mathcal{V}$ be fixed maps, let $\Delta = \{w \in \mathbb{R}^M : w_m \geq 0, \sum_m w_m = 1\}$, and for $w \in \Delta$ write $F_w = \sum_m w_m f_m$. Assume the members' per-sample relative errors are bounded: $\ell(f_m(u), G(u)) \leq B$ for μ-a.e. u and every m.*

(i) For every $w \in \Delta$, $\ell(F_w(u), G(u)) \leq \sum_m w_m \, \ell(f_m(u), G(u))$ pointwise; in particular $R(F_w) \leq \sum_m w_m R(f_m) \leq \max_m R(f_m)$, and $\ell(F_w(u), G(u)) \leq B$.

(ii) Let $\widehat{w}$ minimize the empirical risk $\widehat{R}(w) = \frac{1}{m} \sum_{i=1}^m \ell(F_w(\tilde{u}_i), G(\tilde{u}_i))$ over Δ, computed on an i.i.d. validation sample $\tilde{u}_1, \ldots, \tilde{u}_m \sim \mu$ independent of $f_1, \ldots, f_M$. Then for every $\delta \in (0,1)$, with probability at least $1 - \delta$,

$$R(F_{\widehat{w}}) \;\leq\; \min_{w \in \Delta} R(F_w) \;+\; \frac{8B}{m} \;+\; B\sqrt{\frac{2\big((M-1)\log\big(m(M-1)+1\big) + \log(2/\delta)\big)}{m}}.$$

With $M = 5$ members, $m = 1000$ validation samples and $\delta = 0.05$ the right-hand excess evaluates to about $0.28\,B$; at $B = 0.25$, the largest per-sample member error we observe, this caps the selection cost at roughly seven percentage points. The bound is conservative, as such bounds are: the realized validation-to-test gap of the stack in Section 4 is two orders of magnitude smaller. Its role is the scaling $\sqrt{M \log m / m}$, which says that fitting a handful of convex weights on a thousand held-out samples is statistically almost free; this is why the pipeline spends its validation data there and on the kernel hyperparameters and nowhere else.

The stack the pipeline deploys on the structural-mechanics benchmark is finer than a single convex vector: the weights are affine and vary per grid point, fitted by ridge on the validation split, and the choice between this object and the global convex stack is itself made on held-out data (fit both on one half of the split, keep whichever wins on the other half, refit the winner on the full split). Proposition S1.3 does not cover that construction; the following two statements do, in the same elementary style.

Lemma S1.4 (validated selection). *Let $g_1, \ldots, g_K : \mathcal{U} \to \mathcal{V}$ be fixed maps with $\ell(g_k(u), G(u)) \leq B$ for μ-a.e. u and every k, and let $\widehat{k}$ minimize the empirical risk over an i.i.d. sample of size m' drawn from μ independently of everything used to build the g_k. Then, with probability at least $1 - \delta$,*

$$R(g_{\widehat{k}}) \;\leq\; \min_{k \leq K} R(g_k) \;+\; B\sqrt{\frac{2\log(2K/\delta)}{m'}}.$$

Here $K = 2$ and $m' = 500$: comparing two candidate stacks on half the validation split costs at most a few points of risk in the worst case, and in practice far less. The refit winner is covered by the next statement, which prices the per-coordinate affine fit itself.

Proposition S1.5 (per-pixel affine stacking). *Fix members $f_1, \ldots, f_M$, write $D = \sqrt{M+1}$, and for each output coordinate j let $\varphi_j(u) = (f_1(u)_j, \ldots, f_M(u)_j, b) \in \mathbb{R}^{M+1}$, where b bounds all member and target values: $|G(u)_j| \leq b$, $|f_m(u)_j| \leq b$ for μ-a.e. u and all j, m. Consider the per-coordinate affine class $\{u \mapsto w^\top \varphi_j(u) : \|w\|_2 \leq W\}$ and the coordinate risks $R_j(w) = \mathbb{E}\,(w^\top \varphi_j(u) - G(u)_j)^2$, estimated on an i.i.d. validation sample of size m independent of the members.*

(i) If $\widehat{w}_j$ minimizes the empirical risk of coordinate j over the class, then with probability at least $1-\delta$,

$$\frac{1}{q}\sum_{j=1}^{q}\Big[R_j(\widehat{w}_j)-\min_{\|w\|_2\le W} R_j(w)\Big] \;\le\; \frac{8\,Wb^2D\,(1+WD)+2\,b^2(1+WD)^2\sqrt{\frac{1}{2}\log(4q/\delta)}}{\sqrt{m}}.$$

(ii) If instead $\widehat{w}_j$ minimizes the ridge objective $\widehat{R}_j(w)+\lambda\|w\|_2^2$ and satisfies $\|\widehat{w}_j\|_2\le W$, the same bound holds with an additional λW^2 on the right.

This is a capacity bound, and it should be read as one. Its content is the rate: with a handful of members the per-coordinate fit is paid for by the validation split at rate $\sqrt{(M+1)/m}$ per coordinate, uniformly over coordinates, and the ridge form the pipeline uses adds only its explicit penalty ($\lambda=10^{-3}$ here, so λW^2 is small against the coordinate risks at the fitted weight norms, which the released runs report). Its evaluated value is not informative. The bound is stated for the coordinatewise mean-squared risk in the units of the stress field, not for the relative error of the tables, and the constants b and W enter it polynomially; at the measured values (b is the largest stress value in original units, about a thousand) the right-hand side exceeds the realized excess by many orders of magnitude, as the released evaluation shows. What the statement establishes is that fitting $q(M{+}1)$ numbers on a thousand held-out samples is a selection problem of the ordinary $m^{-1/2}$ kind, with no dependence on q beyond the logarithm; the half-split comparison of Lemma S1.4 then guards the one discrete choice that remains.

One stage of the pipeline is still uncovered at this point: the residual correction, whose scale and nugget are chosen from a finite grid on the same validation split that fitted the stack. The next statement closes the chain. Its key observation is structural: for a fixed grid point the correction is fit to the training residual of the stack, which is affine in the stack weights, so the corrected predictor is again a per-coordinate affine function of w with a transformed feature map, and the machinery of Proposition S1.5 applies to the composite class at the cost of one further constant, a uniform bound on the correction weights, which Lemma S1.7 below supplies in closed form.

Corollary S1.6 (capacity bound for the deployed corrected surrogate)**.** *Assume the setting of Proposition S1.5, and let Γ be a finite set of correction configurations, each defining weights $c_\gamma(u)\in\mathbb{R}^{n_{\mathrm{tr}}}$ with the corrected predictor*

$$h_{w,\gamma}(u)_j \;=\; w_j^\top\varphi_j(u)+c_\gamma(u)^\top\big(Y_j-\Phi_j w_j\big),$$

where $Y_j\in\mathbb{R}^{n_{\mathrm{tr}}}$ collects the training targets of coordinate j and Φ_j the members' training predictions; $\gamma=0$ (no correction, $c_0\equiv 0$) is included in Γ. Suppose the correction weights are uniformly bounded, $\sup_{u,\gamma}\|c_\gamma(u)\|_1\le\kappa$. If the deployed surrogate uses any validation-measurable choice $(\widehat{w},\widehat{\gamma})$ with $\|\widehat{w}_j\|_2\le W$ and $\widehat{\gamma}$ minimizing the empirical validation risk, aggregated over coordinates, among $\gamma\in\Gamma$ at the fitted $\widehat{w}$ (the grid choice is shared across coordinates, as in the pipeline), then with probability at least $1-\delta$,

$$\frac{1}{q}\sum_j R_j\big(h_{\widehat{w},\widehat{\gamma}}\big) \;\le\; \min_{\gamma\in\Gamma}\,\frac{1}{q}\sum_j R_j\big(h_{\widehat{w},\gamma}\big)+\frac{2(1+\kappa)^2\,b^2(1+WD)\Big(4WD+(1+WD)\sqrt{\frac{1}{2}\log\big(4q|\Gamma|/\delta\big)}\Big)}{\sqrt{m}}.$$

Since $\gamma = 0$ is among the candidates, the oracle on the right is at most the fitted stack's own risk, and combining with Proposition S1.5(ii) bounds the deployed risk against the best affine stack itself. The constants b and W are measured and released with the runs; for kernel ridge weights, κ is supplied by the next lemma.

The hypothesis on the correction weights is not an empirical assumption. In the pipeline each $\gamma = (s, \lambda)$ is a length scale and a nugget, and the weights are those of kernel ridge regression, $c_\gamma(u) = (K_\gamma + n\lambda I)^{-1} k_\gamma(X, u)$, so that $\widehat{r}(u) = c_\gamma(u)^\top R$ in the notation of Section 3.3. Their ℓ^1 norm is bounded uniformly in u by the nugget alone.

Lemma S1.7 (uniform bound on kernel ridge weights)**.** *Let k be a positive definite kernel with $k(u,u) \le k_{\max}$ for all u, let $X = (u_1, \ldots, u_n)$ be any design with Gram matrix K, and let $\lambda > 0$. The kernel ridge weights $c_\lambda(u) = (K + n\lambda I)^{-1} k(X, u)$ satisfy, for every u,*

$$\|c_\lambda(u)\|_1 \;\le\; \sqrt{n}\, \|c_\lambda(u)\|_2 \;\le\; \frac{1}{2}\sqrt{\frac{k(u,u)}{\lambda}} \;\le\; \frac{1}{2}\sqrt{\frac{k_{\max}}{\lambda}},$$

and the constant $\frac{1}{2}$ cannot be improved: for every $\lambda \in (0,1)$ there are a kernel, a two-point design and a point u at which the first and last members are equal. In particular, for the residual correction of Section 3.3 — a Matérn kernel with $k(u,u) = 1$ and the nugget grid $\lambda \in \{10^{-6}, 10^{-5}, 10^{-3}\}$ — the hypothesis of Corollary S1.6 holds, whatever the length scale and the design, with

$$\kappa_{\mathrm{an}} \;:=\; \tfrac{1}{2} \max_{\gamma \in \Gamma} \lambda_\gamma^{-1/2} \;=\; 500.$$

The proof is a few lines in Section S8: the Gram matrix of the design together with u is positive semidefinite, which confines $k(X,u)$ to the range of K with $k(X,u)^\top K^{+} k(X,u) \le k(u,u)$, and $\mu/(\mu + n\lambda)^2 \le 1/(4n\lambda)$ for every eigenvalue μ. The bound is the right kind of constant for the corollary: it depends on nothing that is estimated. It is also why the corollary is a capacity bound and not a certificate. The excess term carries $(1+\kappa)^2$, so at $\kappa_{\mathrm{an}} = 500$ the corollary's right-hand side is, like that of Proposition S1.5 before it, vacuous at the shipped constants by many orders of magnitude. The realized scale of the weights is far smaller: on a test probe of five hundred points the largest ℓ^1 norm over the grid averages 32.2 over the ten seeds (Section S10), about a fifteenth of κ_{an}, so that evaluating the display at the probe value would shrink it by a factor of about two hundred and forty. But a supremum over a finite probe is a lower bound on the constant the corollary needs, not the constant, and we do not use it as one. What the corollary establishes is structural: the object the pipeline ships — fitted stack, grid-tuned correction, validated final choice — is covered by an oracle inequality of the ordinary $m^{-1/2}$ form, in the coordinatewise mean-squared metric, whose every constant is either measured or given in closed form. The certificate of this paper, in the sense of a bound whose evaluated value is the quantity reported, is Theorem 6.9.

S2 Decision rules: margins and per-coordinate selection

Part (i) is an exact criterion, but it is used as a decision rule, and a decision rule is applied to estimates. Both ϱ and e_1/e_2 must be formed on data the decision does not see, so what governs the rule in practice is not the criterion but the gap it has to resolve relative to the noise in that estimation. The following makes this quantitative, and is stated for any threshold comparison of this shape.

Proposition S2.1 (margin law for a threshold rule)**.** *Let $D = \varrho - e_1/e_2$ be the signed gap of Proposition 6.1(i), so that mixing the weaker member strictly helps exactly when $D < 0$, and suppose $D \neq 0$. Let $\widehat{D}$ be an estimate of D formed on a split independent of the one on which D is evaluated, and suppose the estimation error $\widehat{D} - D$ is subgaussian with variance proxy σ^2, that is $\mathbb{E} \exp\big(\lambda(\widehat{D} - D)\big) \leq \exp(\lambda^2\sigma^2/2)$ for all $\lambda \in \mathbb{R}$, conditionally on D. Then, writing $\Delta = |D|$ for the margin,*

(i) on $\{D \neq 0\}$, the decision taken from $\widehat{D}$ differs from the correct one with probability at most $\exp(-\Delta^2/2\sigma^2)$*;*

(ii) for every $\tau \geq 0$, and with no condition on D,

$$\mathbb{P}\big(\operatorname{sign} \widehat{D} \neq \operatorname{sign} D, \quad |\widehat{D}| \geq \tau\big) \;\leq\; \exp\left(-\frac{\tau^2}{2\sigma^2}\right).$$

Proof Suppose first $D < 0$, so $\Delta = -D$ and the correct decision is to mix. The decision taken from $\widehat{D}$ differs only if $\widehat{D} \geq 0$, which forces $\widehat{D} - D \geq -D = \Delta$. The subgaussian hypothesis and the Chernoff bound give, for every $\lambda > 0$, $\mathbb{P}(\widehat{D} - D \geq \Delta) \leq \exp(\lambda^2\sigma^2/2 - \lambda\Delta)$, and optimizing at $\lambda = \Delta/\sigma^2$ yields $\exp(-\Delta^2/2\sigma^2)$. If instead $D > 0$ then $\Delta = D$, the decision differs only if $\widehat{D} \leq 0$, which forces $\widehat{D} - D \leq -\Delta$, and the same argument applied to $-(\widehat{D} - D)$, which is subgaussian with the same proxy, gives the identical bound. The two cases are exhaustive because $D \neq 0$, which proves (i).

For (ii), condition on D again. If $D < 0$ the error event is $\{\widehat{D} \geq 0\}$, so on the error event intersected with $\{|\widehat{D}| \geq \tau\}$ we have $\widehat{D} \geq \tau$ and therefore $\widehat{D} - D \geq \tau - D \geq \tau$; the one-sided estimate above applies with Δ replaced by τ. If $D \geq 0$ the error event is $\{\widehat{D} < 0\}$, so on it with $|\widehat{D}| \geq \tau$ we have $\widehat{D} \leq -\tau$ and therefore $D - \widehat{D} \geq \tau + D \geq \tau$, and the one-sided estimate in the other direction applies. In both cases the bound is $\exp(-\tau^2/2\sigma^2)$, which is free of D, so taking the expectation over D removes the conditioning. ■

Three remarks on how the statement is used in Section 5.2. Part (i) is conditional on the true margin Δ, which is not available when the decision is taken; what is available is $|\widehat{D}|$, and part (ii) is the statement about that quantity. It is a joint bound: it limits the probability that a decision is both wrong and taken at an observed margin of at least τ, and it makes that event rare among all decisions once τ exceeds a few multiples of σ. It is not a bound on the error rate among the decisions whose observed margin exceeds τ — that conditional probability is the joint one divided by $\mathbb{P}(|\widehat{D}| \geq \tau)$, on which the proposition says nothing — so the accuracy within a margin bin is an empirical calibration of the rule and not a guarantee, and that is how Section 5.2 reports it. Both bounds are vacuous for margins of order σ and fall off quickly beyond, so a rule of this shape has a band around its own threshold inside which it carries no guarantee, and the band is set by σ rather than chosen. Second, σ is estimable without knowing D: the differences between $\widehat{D}$ and the value of D realized on a held-out block, taken across many pairs, have exactly this scale. Third, $|\widehat{D}|$ is observable at decision time, so a practitioner can report the observed margin next to the verdict; part (ii) says that verdicts reported at a large margin are seldom wrong, and the scoreboard of Section 5.2 says how the realized accuracy varies with that margin.

A second consequence of the same decomposition explains the final OCO-2 surrogate. Evaluation metrics there are diagonal in the output coordinates (the radiance metric is the weighting s_z), and diagonal metrics decouple.

Proposition S2.2 (per-coordinate combination under diagonal metrics)**.** *For weights $v = (v_{mj})$ that may depend on the output coordinate, let $f_v(u)_j = \sum_m v_{mj} f_m(u)_j$. For any diagonal metric with positive weights w,*

$$\mathbb{E}\big\|\mathrm{diag}(w)\big(f_v(u) - G(u)\big)\big\|_2^2 = \sum_j w_j^2 \, \mathbb{E}\big(f_v(u)_j - G(u)_j\big)^2,$$

so the minimizing weights solve q separate problems, one per output coordinate, none of which involves w. The per-coordinate optimum is the same for every diagonal metric, and it weakly dominates every combination whose weights are shared across coordinates, for all such metrics simultaneously.

The practical content: when a problem is scored in several diagonal metrics at once, the combination stage does not need to know which one matters. Fit the best combination coordinate by coordinate on validation and the result serves all of them; only the members need metric-aware training, which is where the weighted mean of Section 5 enters.

The proposition as stated covers quadratic metrics with fixed diagonal weights, while the experiments report relative errors, which weight each sample by its output norm. The decoupling survives that weighting.

Corollary S2.3 (weighted metrics and the reported error)**.** *Let $a : \mathcal{U} \to [0, \infty)$ be measurable with $\mathbb{E}[a(u)\,\|f_v(u) - G(u)\|_2^2] < \infty$. Then, for every diagonal w,*

$$\mathbb{E}\Big[a(u)\,\big\|\mathrm{diag}(w)\big(f_v(u) - G(u)\big)\big\|_2^2\Big] = \sum_j w_j^2 \, \mathbb{E}\Big[a(u)\,\big(f_v(u)_j - G(u)_j\big)^2\Big],$$

so the minimization still splits into q per-coordinate problems, none involving w. Taking $a(u) = \|G(u)\|_2^{-2}$ and $w = \mathbf{1}$ makes the left side the mean squared relative error: the combination fitted coordinate by coordinate with per-sample weights a on validation is simultaneously optimal, within its class, for the squared relative error and for every diagonal reweighting of it. The reported metric, $\mathbb{E}\,\|\rho_{f_v}\|_2$ with ρ the normalized residual of Section 6.2, is controlled through $\mathbb{E}\|\rho\|_2 \le (\mathbb{E}\|\rho\|_2^2)^{1/2}$; the gap between the two sides is the dispersion factor whose measured value stays near 0.93 across the pipelines of this paper.

The experiments run the combination both ways, unweighted and with the per-sample weights of the corollary, and report both.

Proposition S2.2 and its corollary describe the population optimum; the pipeline computes an empirical selection, coordinate by coordinate, on the validation split. The finite-sample cost of that step is priced by the same argument that priced the stacks.

Proposition S2.4 (empirical per-coordinate selection)**.** *Let $f_1, \dots, f_M$ be fixed maps, $a : \mathcal{U} \to [0, A]$ a bounded per-sample weight, and suppose $a(u)\,(f_m(u)_j - G(u)_j)^2 \le L$ for μ-a.e. u, every member m and coordinate j. On a validation sample of size m_v independent of the members, let $\widehat{m}(j)$ minimize the empirical weighted risk of coordinate j over the M members,* *and let the combination take coordinate j from member $\widehat{m}(j)$. Then, with probability at least $1-\delta$, simultaneously for every coordinate,*

$$R_j\big(\widehat{m}(j)\big) \;\le\; \min_{m\le M} R_j(m) \;+\; 2L\sqrt{\frac{\log(2Mq/\delta)}{2m_v}}, \qquad R_j(m)=\mathbb{E}\big[a(u)\big(f_m(u)_j-G(u)_j\big)^2\big],$$

and consequently, for every diagonal metric w at once, $\sum_j w_j^2 R_j(\widehat{m}(j)) \le \sum_j w_j^2 \min_m R_j(m) + 2L\big(\sum_j w_j^2\big)\sqrt{\log(2Mq/\delta)/(2m_v)}$.

On the OCO-2 bands the deployed combination is exactly this estimator ($M=6$ members, $q=40$ coordinates, $m_v=2000$), in the unweighted form and, for the squared relative error, with $a(u)=\|G(u)\|_2^{-2}$; the proposition prices both at the same rate, and the selection cost it bounds is what the seed experiments measure directly.

The corollary is the quantity this paper keeps measuring, at two different levels. Across architecture families on the structural-mechanics benchmark the pairwise correlations exceed 0.86 at member errors near 4.7%, which places the floor at $4.922 \pm 0.056\%$ in the root-mean-square metric it is stated in; Section 4.5 makes the comparison with the deployed pipeline in that metric rather than across the two. Across random seeds of one architecture on the OCO-2 bands the correlations are 0.78, 0.97 and 0.96 at member errors of 4.4%, 16.4% and 8.2%; measured in the squared metric of the proposition itself, over all 45 seed pairs per band, the floors are 4.360%, 17.970% and 9.248% and the realized ten-seed ensembles sit at 4.419%, 17.997% and 9.267% (Section 5). When an ensemble is at its floor the corollary says where improvement cannot come from. The reported tables use the mean rather than the root mean square of $\|\rho_f\|_2$, and the two are not interchangeable: Remark 6.2 gives the relation, the ratio runs from about 0.89 to 0.94 across the objects here, and every comparison in this paper between a quantity read off S and a table entry is made after converting one to the other. The proposition frames the diversity experiment of Section 4.5: what a new member is worth is legible in the row it adds to S, before any weights are fitted. On this benchmark the rows all look alike, pairwise correlations of 0.83–0.97 among trained networks of three architecture families and two losses, 0.86–0.88 against the kernel predictor, so the floor sits just below the best single member, and the measured stack lands on it. Part (i) is the same computation specialized to two members; it correctly predicts, from measured (e_1, e_2, ϱ) alone, which members receive weight in the fitted stack and which are dropped. Section 5 applies the same test on a problem where the answer comes out the other way: there the kernel's error is inflated by a factor the next subsection quantifies, the threshold e_1/e_2 collapses, and the kernel is dropped even at a residual correlation of 0.14.

Suppose the operator factors as $G(u)=g(Au)$ with $A\in\mathbb{R}^{d\times d}$ nonsingular and g of unit H^s norm, and that μ has a density bounded above and below on a compact domain. Order the singular values $\sigma_1\ge\cdots\ge\sigma_d$ of A; they are the rates at which G varies along the input directions. Suppose A has *approximate rank* r at the sample size, meaning the trailing singular values fall below the achievable resolution, $\sigma_{r+1}\lesssim n^{-1/r}$, so the image $A\mathcal{U}$ is effectively r-dimensional at scale $n^{-1/r}$. The idealization beneath the display also takes the leading-r projection of the image design to keep a density bounded above and below, the trailing directions to be exact zeros at the working resolution, and the interpolation limit $\lambda\to 0$; Supplement S8 records the argument at that level, as a sketch rather than a complete proof. The statement is accordingly a heuristic rate calculation, and it is labelled informal: nothing else in the paper rests on it, and its role is to say what kind of gain a metric can and cannot produce.

Proposition S2.5 (isotropic and adapted kernel rates, informal)**.** *Let $\widehat{G}_n^{\mathrm{iso}}$ be kernel ridge regression with a Matérn kernel of smoothness s and a validation-optimal single length scale, and $\widehat{G}_n^{\mathrm{ard}}$ the same construction in the metric $u \mapsto Au$. Up to constants depending on (σ_i), g, s and the density bounds, and up to logarithmic factors in n, with high probability over n i.i.d. samples,*

$$\mathbb{E}\big\|G - \widehat{G}_n^{\mathrm{iso}}\big\| \;\lesssim\; \sigma_1^s\, n^{-s/d}, \qquad \mathbb{E}\big\|G - \widehat{G}_n^{\mathrm{ard}}\big\| \;\lesssim\; n^{-s/r},$$

so the ratio of the two bounds is of order $\sigma_1^s\, n^{\,s(1/r-1/d)}$. When A is close to full rank ($r \approx d$) the two rates coincide and only the constant σ_1^s separates them.

The calculation (Supplement S8) is the scattered-data estimate $\|G - \widehat{G}_n\| \lesssim h_X^s \|G\|_{H^s}$ applied to the two designs: one length scale must fill all d directions, so $h_X \asymp n^{-1/d}$ and the norm carries σ_1^s; the adapted metric, when A has a spectral gap, confines the design to r resolved directions and improves the fill distance. Both displays are upper bounds under the stated idealization — the exact metric A, a fixed g, and the length-scale selection folded into the logarithmic factors — and we claim no matching lower bound; the heuristic is used here to calibrate what a metric can and cannot buy, not as a rate theorem. It isolates the *metric* part of a network's advantage, the part an anisotropic kernel would recover, and it says this part is a rate gain only under an approximate-rank gap; without one it is the constant σ_1^s alone. The rest is *representational*: a stationary kernel of fixed smoothness reaches only its native space, at any metric, while the network adapts its features to the map. The two parts are separated experimentally by handing the kernel the anisotropic metric and seeing how much of the gap closes. On the OCO-2 bands of Section 5 the metric closes a factor of 1.1 to 1.5 out of a tenfold gap (Section 5.1); the remainder is representational, and its direct evidence is that the same kernel machinery applied to the network's learned features recovers the network's accuracy and slightly exceeds it.

The rate half of the heuristic is tested rather than assumed, and the test is the more interesting for not going our way. Both kernel rows are refit at nested training sizes over a sixteenfold range, at ten seeds per band. The sensitivity map's approximate rank is 0.54 to 0.60 of the input dimension, a gap wide enough that the calculation would permit a rate gain; the measured difference between the two empirical exponents is nevertheless small on all three bands and consistent with zero on one. The displays are upper bounds with no matching lower bound, so this does not contradict them — the adapted rate is free to beat its own bound — but it does mean that on these problems the rank gap is not converted into a rate, and a reader should take from Proposition S2.5 only the part the measurement supports: an anisotropic metric is worth a constant. The controlled rank-gap target of the verification suite, where the construction is exactly the idealization the calculation assumes, does show the adapted exponent improving; the distance between that and the OCO-2 bands is the distance between the idealization and a real sensitivity map.

S3 Distribution-free coverage for the reported band

S3.1 Distribution-free coverage for the reported band

Theorem 6.9 controls the error by $\|G - m\|_K \, P_\lambda(u)$, but the constant $\|G - m\|_K$ is not known a priori, and Section 4.7 shows that P_λ alone ranks the error weakly. What the surrogate reports as uncertainty is therefore not P_λ itself but a rescaling of it, $q\, P_\lambda(u)$, whose multiplier q is calibrated on the held-out split by the split-conformal rule

$$q = Q_{1-\alpha}\Big(\big\{s_i := \|e(\tilde{u}_i)\|_2 / P_\lambda(\tilde{u}_i)\big\}_{i=1}^{m}\Big), \qquad e(u) = G(u) - m(u) - \widehat{r}_\lambda(u), \tag{S3.1}$$

where $Q_{1-\alpha}$ is the $\lceil (1-\alpha)(m+1) \rceil$-th smallest value of the m validation scores. The point of this construction is that its coverage needs neither the bound to be tight nor P_λ to be a good error ranking; it needs only exchangeability, which the fixed splits provide.

Proposition S3.1 (finite-sample coverage)**.** *Fix the mean m, the correction $\widehat{r}_\lambda$ and the design X, and let $P_\lambda(\cdot) > 0$. Suppose the validation and test inputs $\tilde{u}_1, \ldots, \tilde{u}_m, u^\star$ are exchangeable (in particular i.i.d. from μ) and drawn independently of $m, \widehat{r}_\lambda, X$. Let q be the conformal multiplier* (S3.1) *and define the band $C(u) = \{v : \|v - m(u) - \widehat{r}_\lambda(u)\|_2 \le q\, P_\lambda(u)\}$. Then*

$$1 - \alpha \;\le\; \mathbb{P}\big(G(u^\star) \in C(u^\star)\big) \;\le\; 1 - \alpha + \frac{1}{m+1},$$

the upper bound holding when the scores are almost surely distinct.

Remark S3.2 (the exact law of realized coverage)**.** *When the calibration and test scores are i.i.d. with a continuous distribution F — as in the campaign's protocol, where calibration and evaluation sets are a random split of an i.i.d. test block — the coverage probability conditional on the calibration sample is F evaluated at the k-th smallest calibration score, $k = \lceil (1-\alpha)(m+1) \rceil$. Since $F(s_i)$ are i.i.d. uniform, this is the k-th order statistic of m uniforms, with distribution exactly* $\mathrm{Beta}(k,\, m+1-k)$ *(Vovk, 2012); its mean $k/(m+1)$ recovers Proposition S3.1. Conditional on the calibration sample the evaluation indicators are i.i.d. Bernoulli, so the coverage observed on N evaluation points follows the Beta–Binomial law with these parameters. Each seed's observed coverage is therefore placed against the central band of an exact finite-sample law, whose per-seed width at* $m = 1000$ *is close to $\sqrt{\alpha(1-\alpha)/m}$.*

The two-sided statement is the standard guarantee of split conformal prediction (Vovk et al., 2005; Lei et al., 2018), transported to the functional-output setting by taking the nonconformity score to be the relative residual norm $s = \|e(u)\|_2 / P_\lambda(u)$; the proof, a rank argument on the exchangeable scores, is in Supplement S8. Three points make it the right closing statement for the uncertainty analysis. It is close to the quantity we compute: the reported coverage of $90.2 \pm 0.6\%$ at the nominal $1 - \alpha = 90\%$ in Section 4.7 realizes this proposition with $m = 1000$ and $\alpha = 0.1$. Any gap to nominal is not the $1/(m+1)$ slack, which is only a tenth of a point; it is the finite-sample fluctuation of conditional coverage for a single calibration draw, whose standard deviation $\sqrt{\alpha(1-\alpha)/m}$ is about a point at these sizes. Remark S3.2 gives the exact law of that fluctuation, and the ten seeds let it be checked rather than invoked: the observed spread across seeds is 0.6 points against the 0.9 points the law predicts, and every seed falls inside the central 95% band of the exact distribution at the 90% level and at the 95% level alike. The hypotheses hold exactly as stated: calibration uses a random subset of the test block that no training, checkpointing, stacking, or tuning step ever touched, and coverage is evaluated on the disjoint remainder (Supplement S9), so the exchangeability the proposition assumes is exact, with no low-complexity selection argument needed. It is agnostic to everything the earlier results leave uncertain: the neural mean may be biased, P_λ may correlate with the error weakly or with the wrong sign, and the bound of Theorem 6.9 may be loose, yet the band still covers at the stated rate. And it is where the uncertainty analysis ends: among the candidate uncertainty signals we examine, the conformally rescaled P_λ is the only one that carries a guarantee, so it is the one the surrogate reports.

S4 The rest of the suite, and the mirror-symmetry check

S4.1 The remaining problems of the suite: an uncontrolled survey

For orientation we also ran one fixed recipe, unchanged, on the other problems of the operator-learning suite that Batlle et al. (2024) assembled from de Hoop et al. (2022) and Lu et al. (2022): a Matérn kernel at the median length scale with the fit capped at 6000 points, a residual MLP mean, outputs reduced by PCA, and the stage selected on validation. Table S4.1 lists the results next to the best published number on each problem. The comparison is not controlled, and we draw nothing from it beyond orientation: the published numbers are per-problem methods tuned by their authors; our rows are single runs without seed replication, except Helmholtz and Navier–Stokes, which were rerun at three seeds with the exact solve on 19,000 training pairs once it was found that the recipe's output rank of 40, adequate for the stress fields, left a reconstruction floor of 1.7% and 8.2% on these two problems (the rerun uses rank 400, whose floor is below 0.01%, and every rerun head sits well above it); Burgers and Darcy use slightly smaller training splits than their baselines (800 and 824 pairs) and Darcy a different grid, from the copies we could obtain; and we report no number for Advection I. What the survey shows is the sorting one would expect: where the map is smooth and data are plentiful (Helmholtz, Navier–Stokes, Darcy) the kernel stage wins the recipe's internal selection and lands within a factor of 1.2 (Helmholtz) to 2.3 (Navier–Stokes) of the tuned per-problem kernels, whose Cholesky preconditioning of the input the recipe does not use, and on Burgers the corrected mean sits near the Fourier neural operator. On Helmholtz the per-coordinate selection improves on the kernel alone (1.17% against 1.22%) by handing a few dozen of the 400 output coordinates to the corrected mean; on Navier–Stokes it selects the kernel almost everywhere. The two problems studied in depth, with replication, are the subject of the rest of the paper.

S4.2 A data-driven check of the mirror symmetry

The continuous problem is invariant under $x_1 \mapsto 1 - x_1$: the domain and boundary partition are symmetric, and the input law has a symmetric covariance. On the grid, reflecting the load (S) should reflect the stress field along the first grid coordinate (T), i.e. $G(Su) = TG(u)$. Rather than assume this, we test it on the data. For each of the first 200 samples we searched all 40000 loads for the nearest neighbor of the reflected load Su_i and compared the corresponding outputs. For the five best-matching pairs, the input mismatch $\|Su_i - u_j\|/\|u_j\|$

Table S4.1: The remaining problems of the suite: an uncontrolled survey. Best published mean relative L^2 test error on each problem (per-problem methods tuned by their authors), against one recipe run unchanged, on training splits and grids that differ from the baselines' in the cases noted in the text: single runs, except the two rows marked three seeds (mean ± standard deviation, exact solve on 19,000 pairs, output rank 400).

Problem	Map	Best published	This work
Burgers	initial condition → solution	1.93% (FNO)	2.38%
Darcy flow	permeability → pressure	2.32% (POD-DeepONet)	2.01%
Advection I	smooth pulse → solution	≈ 0% (kernel)	—
Advection II	discontinuous pulse → solution	11.28% (linear kernel)	11.29%
Helmholtz	wavespeed → field	1.00% (kernel)	1.17% ± 0.01 (three seeds)
Navier–Stokes	vorticity → vorticity	0.12% (kernel)	0.28% ± 0.00 (three seeds)

ranges over 0.27–0.32, while the output mismatch $\|Tv_i - v_j\|/\|v_j\|$ ranges over 0.085–0.14; reflecting along the second coordinate instead gives output mismatches of order one. Outputs of near-mirror inputs are far closer than the inputs themselves, which is what equivariance plus a Lipschitz solution map predicts, and the effect singles out the correct output reflection axis. A complementary check on the input law: the empirical mean and covariance of the training loads are invariant under reflection to within 1.1% and 1.5% respectively, as required for Proposition S1.1. Consistently with all of this, reflection averaging at test time improves twenty-nine of the thirty reflected members of the seed campaign (Section 4), which is what Proposition S1.1 predicts: it constrains the expected loss, so a finite test sample is free to go the other way, and at one seed it does.

S5 Effective dimension from the Gram spectrum

S5.1 Effective dimension from the Gram spectrum

The remaining design choice is the nugget λ, selected by cross-validation in the pipeline. The following exact identity connects that choice to the spectrum of the Gram matrix and, through it, to random matrix descriptions of the design.

Lemma S5.1 (effective dimension and the empirical Stieltjes transform)**.** *Let K be a symmetric positive semidefinite $n \times n$ matrix with eigenvalues $\lambda_1, \dots, \lambda_n \geq 0$, let $\widehat{m}(z) = \frac{1}{n}\sum_{i=1}^{n}(\lambda_i/n - z)^{-1}$ be the Stieltjes transform of the empirical spectral distribution of K/n. Then for every $\lambda > 0$ the effective dimension of ridge regression with nugget λ satisfies*

$$d_{\mathrm{eff}}(\lambda) \;:=\; \operatorname{tr}\big(K(K + n\lambda I)^{-1}\big) \;=\; n\big(1 - \lambda\,\widehat{m}(-\lambda)\big).$$

In particular, if the empirical spectral distribution of K/n converges weakly to a limit law F as $n \to \infty$, then $d_{\mathrm{eff}}(\lambda)/n \to 1 - \lambda\, m_F(-\lambda)$ with m_F the Stieltjes transform of F.

The identity is unconditional; the random-matrix content enters through the choice of F. For the benchmark's simplest reference model, isotropic linear features, the limit can be evaluated in closed form.

Corollary S5.2 (effective dimension under a Marčenko–Pastur limit). *Let $K = XX^\top$ with $X \in \mathbb{R}^{n\times p}$ having i.i.d. entries of mean zero and variance σ^2, and let $p/n \to \gamma \in (0,1]$. Then*

$$\frac{d_{\mathrm{eff}}(\lambda)}{n} \longrightarrow \gamma\big(1-\lambda\, m(-\lambda)\big), \qquad m(-\lambda) = \frac{\sqrt{\big(\lambda+\sigma^2(1-\gamma)\big)^2 + 4\gamma\sigma^2\lambda} - \big(\lambda+\sigma^2(1-\gamma)\big)}{2\gamma\sigma^2\lambda},$$

almost surely, where m is the Stieltjes transform of the Marčenko–Pastur law with ratio γ and scale σ^2. In a simulation with $n = 3000$, $p = 900$, $\sigma^2 = 1.7$, the formula agrees with the empirical $d_{\mathrm{eff}}(\lambda)/n$ to four decimal places across $\lambda \in [10^{-3}, 10]$.

Two uses. Quantitatively, the formula makes the nugget–capacity trade-off explicit for the bulk of a feature Gram matrix: eigenvalue mass of Marčenko–Pastur type is absorbed or discarded by λ according to a single closed-form curve, and outlying (spiked) eigenvalues x_s simply add their terms $x_s/(x_s+n\lambda)$ on top. Qualitatively, it separates the two regimes visible in Figure S6.3: the Matérn Gram matrix on the loads is nothing like a Marčenko–Pastur bulk (its spectrum decays exponentially, which is why d_{eff} is a few hundred out of 19000), whereas the Gram matrices of learned penultimate features do show a bulk-plus-spikes shape, and for them the corollary describes how much of the bulk the cross-validated nugget retains. For the feature Gram matrices of the trained networks we observe the familiar picture of a bulk that is well fitted by a Marčenko–Pastur law (Marčenko and Pastur, 1967) together with a small number of outlying eigenvalues carrying the regression signal, as in spiked covariance models (Baik et al., 2005); for the Matérn Gram matrix on the 41-dimensional loads the spectrum decays rapidly and d_{eff} is small ($\approx$ a few hundred at the cross-validated λ, out of $n = 19000$; Figure S6.3). This gives a post-hoc reading of the cross-validated nugget: λ lands where $d_{\mathrm{eff}}(\lambda)$ has absorbed the outlying eigenvalues and the leading bulk, and further decreasing λ buys capacity precisely where the spectrum carries little signal. It also explains why the exact solve is affordable: the correction is numerically a problem of dimension d_{eff}, not n.

The effective dimension is also exactly the total budget of the design factor from Theorem 6.9. Writing $A = K + n\lambda I$, the posterior variances at the training inputs sum to

$$\sum_{i=1}^{n} P_\lambda(u_i)^2 = \mathrm{tr}(K) - \mathrm{tr}(KA^{-1}K) = \mathrm{tr}\big(n\lambda\, KA^{-1}\big) = n\lambda\, d_{\mathrm{eff}}(\lambda),$$

using $KA^{-1}K = K - n\lambda\, KA^{-1}$. So the same d_{eff} that reads the nugget off the spectrum also fixes, up to the factor $n\lambda$, the total posterior-variance mass the correction distributes over the design; the two halves of this section are one quantity seen from two sides.

S6 Additional structural-mechanics results

S6.1 Ablation

Table S6.1 builds the pipeline up one component at a time. The kernel ridge on loads and the reflection-averaged neural mean start at 5.197% and 4.836%. Stacking is trivial with a single high-data mean and leaves it unchanged; the value of the combination appears once there are several means to combine and again in the correction stage. The feature-space and second input-space corrections did not improve validation error beyond the first correction

Table S6.1: Building the high-data pipeline. Mean relative L^2 test error under the 20000-sample protocol; "+" rows are cumulative. All rows are mean $\pm$ standard deviation over ten seeds; the last two are the second campaign of Section 4.1.

Stage	Test error (%)
Kernel ridge on loads	5.197 ± 0.003
Reflection-averaged neural mean	4.836 ± 0.018
+ refiner and MSE variant, affine stack	4.628 ± 0.009
+ residual kernel correction	4.611 ± 0.005
+ FNO member, affine stack	4.595 ± 0.008
+ residual kernel correction	4.572 ± 0.010
+ complete FNO schedule and UNet member, affine stack	4.567 ± 0.004
+ residual kernel correction	4.546 ± 0.003

with a single mean and so were not selected; we report the stages that the validation split actually chose.

Two smaller effects are worth isolating. Reflection test-time averaging lowers the metric-trained MLP's test error by 0.16 points, consistent in direction with Proposition S1.1, which fixes the sign of the expected change but not its size; the effect is much smaller on the normalized-MSE network (0.02 points) and absent by construction from the refiner, which trains symmetrized. Adding the kernel-flow regularizer (Section S1.2) to the low-data MLP did not help and was not selected; Proposition S1.2 says what the term estimates, not that estimating it is useful on every problem, and on this smooth-output benchmark the plain metric loss was already a good proxy. Nor does a richer correction kernel help: replacing the single tuned Matérn with a sum of Matérn kernels at several bandwidths leaves the validation error unchanged to two decimals, which is consistent with the residual being close to kernel-irreducible in the input geometry, and is why the boosted correction stages of Section 3.3 were not selected. The correction's own grid confirms the same reading from the other side: at all ten seeds the tuner selected the largest nugget on its grid, $\lambda = 10^{-3}$, at nine of the ten (the tenth took 10^{-5}), and a length scale of twice the median at five, of the median itself at four and of four times it at one — the most heavily regularized correction available to it, chosen independently at nearly every seed. The length scale is the less stable of the two choices, and the four-member ablation concentrates more tightly than this (eight seeds at twice the median), so the spread here belongs to the five-member configuration.

The refiner's conditioning field is the kernel prediction, and the choice is not incidental: of all the members, the kernel's residual is the least correlated with the networks' (Section 4.5), so the field it supplies carries information complementary to the network's inductive bias in a way a second network's field would not. Enlarging the ensemble beyond the residual family is the subject of Section 4.5, where the returns of each addition turn out to be predictable in advance from the correlation structure of the members' errors.

S6.2 Where the remaining error is

The seed replicates make it possible to ask not just how large the error is but where it sits, and whether the ensemble touches it. Both questions have the same answer at every seed.

Per sample, the error is concentrated and the concentration is not where an ensemble can help. Ranked by relative error on the validation split, the twenty hardest cases sit at 10.24% against a median of 4.44%, a factor of 2.31. On exactly those cases the five-member ensemble reaches 10.18% and the mean over the individual members is 10.28%: averaging five members, including the architecturally distinct one, recovers a tenth of a point against the average member and six hundredths against the best one, on cases whose error is more than twice the median. Everywhere else the ensemble buys what Section 4.5 says it should; on the hard tail it buys almost nothing, because the members fail there together — and adding the spectral member, which helps everywhere else, does not change that. Within this ensemble, then, the hard tail is not a shortage of member diversity; whether a surrogate class we did not train would fail there too is not something five architectures can settle, but the location of these cases, at the loaded edge and the supports, is where the shared residual of Section 4.5 concentrates.

Spatially, the error is a boundary phenomenon. The root-mean-square pointwise error over the validation samples is 8.99 on the perimeter of the 41×41 grid against 5.64 in the interior, a ratio of 1.59; both figures are stable across seeds to under a percent of themselves. Averaging lowers them to 8.87 and 5.59, so the ensemble takes about a tenth off a boundary error of nine and half that off an interior error of five and a half: it shaves the concentration slightly but does not touch its shape. The location agrees with the anatomy of the shared component above and with the error maps of Figure 1: it is the loaded edge and the supports, where the traction condition meets the geometry, not a diffuse modeling deficit.

S6.3 Spectra and effective dimension

Figure S6.3 shows the spectrum of the Matérn Gram matrix on the loads and the effective dimension $d_{\text{eff}}(\lambda)$ computed from it by the exact identity of Lemma S5.1. The spectrum decays quickly: a few hundred eigenvalues carry essentially all the mass, the tail falling many orders of magnitude below the leading modes. At the cross-validated nugget the effective dimension is a few hundred out of $n = 19000$. This is a statistical statement, not a computational one — the dense factorization costs its full n^3 operations regardless — but it is why the correction generalizes despite interpolating nineteen thousand points: the problem the kernel solves has the statistical dimension of the retained spectrum, not of the sample. The cross-validated λ sits at the shoulder of the d_{eff} curve, past the leading modes and into the rapidly decaying tail, matching the reading of Lemma S5.1.

S6.4 Cost and reproducibility

The pipeline was built on a single laptop and replicated on four cloud CPU instances. The kernel stages are exact: one Cholesky factorization of the 19000×19000 Matérn Gram matrix in double precision takes 24 seconds on the laptop's CPU (16 threads, OpenBLAS; the matrix itself is 2.9 GB), and between 18 and 32 seconds on the campaign instances, on which it ran ten times, once per seed. Prediction is two matrix products; there are

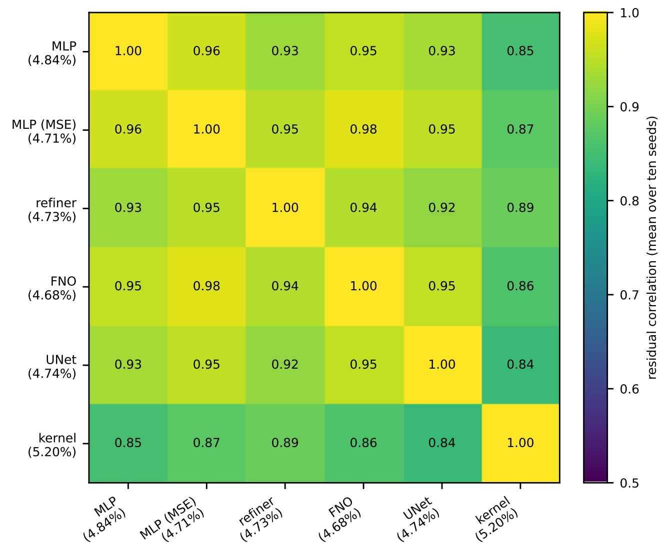

Figure S6.1: Correlation of per-sample normalized residuals on the validation split, six members, averaged over the ten seeds of the second campaign. Across seeds and pairs the entries range between 0.82 and 0.98, with a mean of 0.919 ± 0.005; the kernel predictor, the most alien member by construction, still correlates 0.86–0.88 with the most accurate network at every seed.

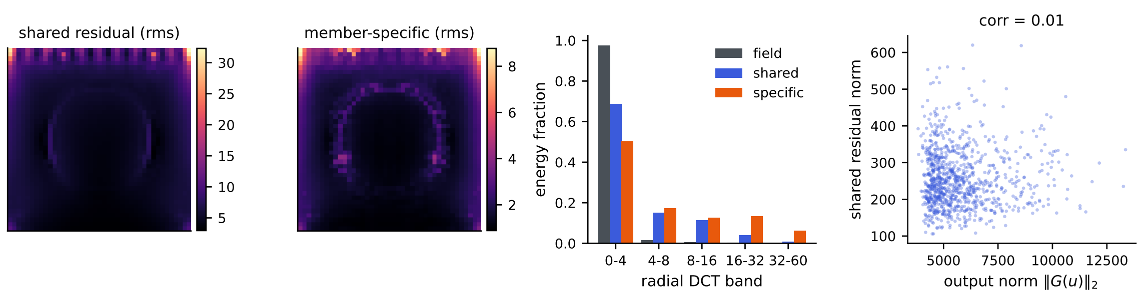

Figure S6.2: Anatomy of the shared residual c (across-member mean of the residual fields), at one seed. Left to right: rms of c over the grid; rms of the member-specific remainders; radial DCT band energies of the stress fields, of c, and of the specific parts; and the norm of c against the output norm, whose correlation is 0.01.

no inducing points, stochastic solvers, or preconditioners, and no GPU is needed for the correction. The factorization performs the full $n^3/3 \approx 2.3 \times 10^{12}$ floating-point operations — numerical low rank does not reduce the work of a dense Cholesky — and the measured rate, about 100 gigaflops per second, is simply what a current multicore CPU delivers; exactness at this n costs half a minute, not a cluster. What the low effective dimension of Section S6.3 buys is statistical rather than computational: it describes where the solve's information concentrates at the working nugget, not its flop count.

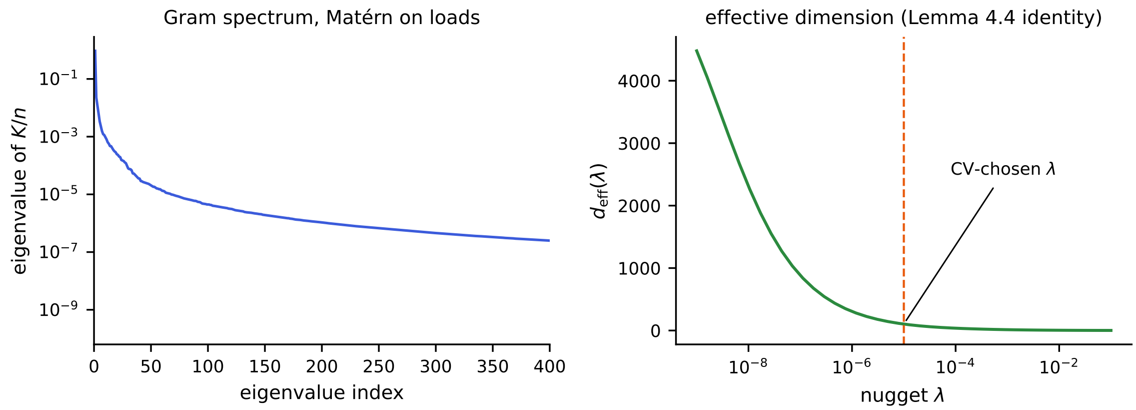

Figure S6.3: Left: eigenvalues of the Matérn Gram matrix K/n on the loads (log scale); the spectrum decays by many orders of magnitude within a few hundred modes. Right: the effective dimension $d_{\text{eff}}(\lambda)$ from the exact identity of Lemma S5.1, with the cross-validated nugget marked.

Building the Gram matrix, not factoring it, is the stage that needs care at this size: forming it through a full squared-distance matrix and its elementwise transforms allocates several 2.9 GB temporaries at once, which exhausted a 31 GB instance during the campaign. Filling it in row blocks into a single preallocated array holds the peak at the matrix itself and is what the released code does. The neural means are small by the standards of the literature (a few million parameters) and train in the metric itself. Every split is fixed by a published permutation seed, the input is consumed as the 41-dimensional load after the exact broadcast check of Section 2, and every model is selected on the validation split and evaluated once on the test set; we release the splits, the code, and the per-seed result record behind every stage of every run, so any row of any table can be rechecked against the records or regenerated end to end by rerunning the pipeline. The prediction arrays and checkpoints themselves are not distributed, so regenerating a row means retraining it.

S6.5 Uncertainty quantification

Theorem 6.9 certifies the corrected surrogate's error by $\|G - m\|_K \, P_\lambda(u)$, and it is tempting to read the design factor P_λ as a pointwise error bar. On this benchmark that reading fails, in an instructive way. Table S6.2 shows the operator factor moving in the direction the theory suggests: the squared RKHS norm of the fitted correction is about $271\times$ smaller when the kernel regresses ensemble residuals than when it regresses raw stress fields. The norm of a fitted interpolant is a lower bound on the norm of the target it interpolates, so the ratio compares two lower bounds and is read as a diagnostic of the fitted objects rather than as a bound on the constant of Theorem 6.9. The design factor P_λ correlates only weakly with the corrected surrogate's absolute error (Pearson 0.08), and *negatively* with the benchmark's relative error. The cause is measurable: P_λ is large for loads far from the training set, those loads tend to have large amplitude, large-amplitude loads produce large-norm stress fields, and dividing by the output norm makes their relative error small.

Indeed P_λ correlates 0.65 with the output norm (Figure S6.4). The error is dominated by the neural mean, whose mistakes are not a function of the input-space geometry that P_λ sees, so a purely input-space power function cannot rank them.

The obvious alternative fares no better, and it fails in the same direction, which is the part worth noticing. Ensemble disagreement, the spread of the members' predictions, is the standard deep-ensemble error signal. Its correlation with the corrected surrogate's absolute error is 0.074 ± 0.006 over the ten seeds, and with the benchmark's relative error it is -0.25 ± 0.04: weakly positive against the quantity the theory speaks about, negative against the quantity the benchmark reports, exactly as for P_λ, and for the same amplitude reason. Two uncertainty signals of quite different provenance — one a Gaussian-process design factor computed from the inputs alone, the other a spread across trained models — fail on this problem in the same way and to the same degree. Adding the spectral member does not repair it: the same correlation is 0.083 on the four-member ensemble, so the member whose admission is worth 0.039 points of accuracy buys nothing as an uncertainty signal. Section 4.5 says why it cannot: disagreement measures the member-specific components of the error, which carry under a tenth of the residual energy, while the ranking signal for the shared component is invisible to any within-ensemble statistic, because the members agree precisely where they are jointly wrong. Section S6.2 is the same fact seen on the hard tail.

What does hold is coverage, and it holds at every seed. We take the exact split-conformal quantile of the corrected surrogate's error score on a calibration subset drawn from the test block — 1000 calibration cases against 19000 evaluation cases, neither used for training, checkpointing, stacking, or kernel tuning — and evaluate coverage on the disjoint remainder. Conformal validity needs no correlation between the score and the error, only exchangeability, which this protocol supplies exactly (Section 6.5). At the nominal 90% level the observed coverage is $90.2 \pm 0.6\%$, and the right comparison is not with 90% but with the exact finite-sample law: with $m = 1000$ calibration points the coverage of a split-conformal band is Beta$(901, 100)$ distributed, whose central 95% interval is $[88.1\%, 91.8\%]$. All ten seeds fall inside it. At the 95% level the band is $[93.6\%, 96.3\%]$ and the observed coverage is $95.0 \pm 0.6\%$, again with all ten inside. The six-member pipeline gives $90.1 \pm 0.4\%$ and $95.0 \pm 0.8\%$, with all ten seeds inside at the 90% level and nine at the 95% level, the tenth on the band's upper edge. Scaling the score by ensemble disagreement, which the correlations above say should not help, indeed does not: it changes mean coverage by three tenths of a point and leaves every seed inside the same band. The lesson is narrow but real: a valid pointwise bound (Theorem 6.9) is not automatically a useful pointwise indicator, the standard uncertainty signals can both fail on the same problem, and a relative-error metric has to have its normalization accounted for before a posterior standard deviation means what it appears to.

S6.6 Which uncertainty is the binding one

Seeds are not the only source of uncertainty in Table 2, and on this benchmark they are not the largest. Every seed is scored on the same fixed 20000-sample test block, so the seed standard deviation measures training and selection variability and nothing else. The test block is itself a sample. Its contribution is the standard error of the mean relative error over those 20000 cases, and at a per-sample standard deviation of 0.0169 that is 0.012 points,

Table S6.2: Squared RKHS norm of the fitted kernel interpolant, $\mathrm{tr}(\alpha^\top K\alpha)$, when the same validated kernel and design regress raw stress fields versus ensemble residuals.

Kernel regresses	$\|\widehat{r}\|_K^2$
Raw stress fields ($m = 0$)	1.39×10^{10}
Ensemble residuals	5.14×10^{7}
Ratio	$271\times$

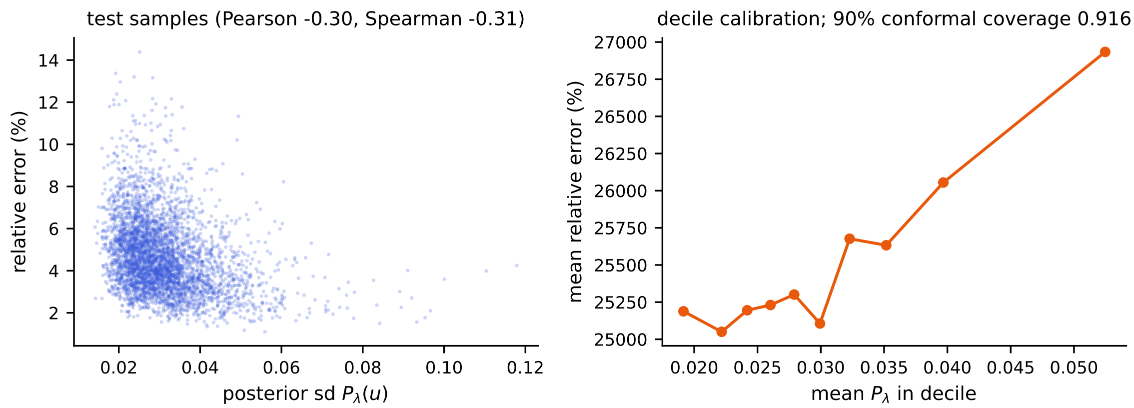

Figure S6.4: Both panels are one seed. Left: the Gaussian process posterior standard deviation P_λ against the corrected surrogate's relative error on the test set; the association is weak and, because of the output-amplitude confound, slightly negative. Right: decile calibration of P_λ against absolute error, the theory-consistent quantity. Across all ten seeds the distribution-free conformal band attains $90.2 \pm 0.6\%$ coverage at the nominal 90% level, inside the exact finite-sample interval at ten seeds out of ten.

slightly larger than the pipeline's seed spread of 0.010. Treating the two as independent, the uncertainty on our reported 4.572% is about 0.016 points, and the seeds contribute a little under half of it. A study that replicates seeds and stops has measured the smaller half and is liable to quote it as though it were the whole.

This decides what the table can and cannot support. Comparisons against a published number are unpaired, because we do not hold those methods' per-sample predictions, and so carry the test-block error twice: about 0.020 points in quadrature. The separation from PCA-Net, 0.098 points, is five of those and real. The separation from PARA-Net is 0.022 points, about one, and is therefore not a separation at all. Both figures are conservative, since scoring on identical test points makes the two errors positively correlated in a way we cannot exploit without the baselines' predictions.

Comparisons inside the pipeline are paired, and pairing is worth far more here than seeds are. Evaluating the corrected surrogate and the best single member on the same cases removes the shared difficulty of each case, and the gain of 0.140 points is then resolved at roughly a hundred standard errors at every seed. The same holds for the FNO's admission: 0.039 points, paired, is overwhelming on any one seed, while the interesting quantity —

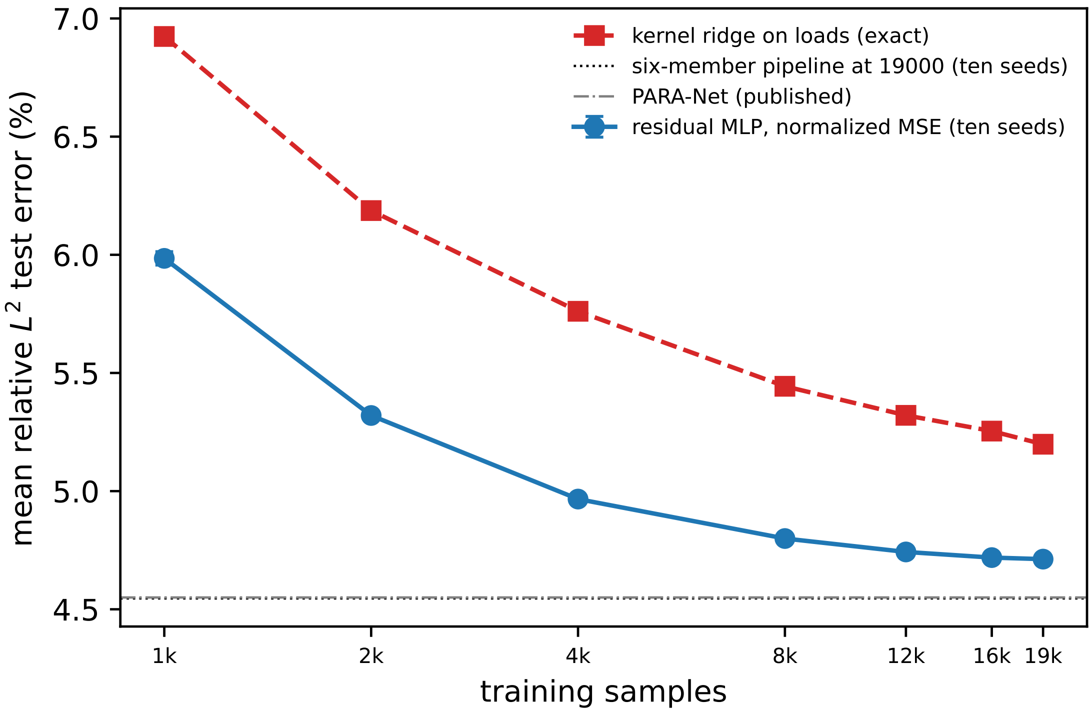

Figure S6.5: Learning curves at ten seeds under one protocol (the first N rows of the training pool, a fixed 1000-case validation carve, the full test block): mean relative test error of the normalized-MSE network, mean and standard deviation over ten seeds, and of the deterministic Matérn kernel ridge on loads, against the number of training samples, both axes logarithmic. The horizontal lines mark the six-member pipeline at 19000 samples and the best published result. Error bars are smaller than the markers above 4000.

whether it survives retraining — is the across-seed spread of ± 0.011 and the count of ten seeds out of ten. The practical rule the budget implies is the ordinary one, and the campaign is what made it visible: replicate seeds to learn whether a design choice survives retraining, pair on the test set to learn whether it helps, and never use either number to answer the other's question.

The weights it assigns are as informative as the risk it predicts, and they correct something. The optimum spreads across all five members at essentially every seed — refiner 0.31 ± 0.04, FNO 0.34 ± 0.06, normalized-MSE network 0.24 ± 0.08, kernel 0.06 ± 0.03, metric-trained MLP 0.06 ± 0.04 — with the kernel receiving a nonzero share at ten seeds out of ten and the plain MLP at nine. Solved properly the answer is a spread, not a vertex: every member earns some weight. On the six members of the second campaign the shares are UNet 0.32 ± 0.06, FNO 0.30 ± 0.07, refiner 0.27 ± 0.04, kernel 0.06 ± 0.02, normalized-MSE network 0.03 ± 0.04 and metric-trained MLP 0.02 ± 0.02.

That the FNO takes the largest single share while ranking only third on the metric is the whole content of the diversity argument. S is a matrix of second moments, so what it rewards is a residual that is both small and unlike the others', and the spectral member is the only architecture here that is structurally unlike the residual-MLP family. The pairwise admission test of part (i) — a member of error e_2 helps a reference of error e_1 when their correlation falls below e_1/e_2 — admits it at all ten seeds, though narrowly: correlation 0.95–0.97 against a threshold of 0.987–1.001. The kernel is admitted more comfortably, 0.86–0.88 against 0.905–0.918, and also enters the deployed pipeline twice more, through the refiner it conditions and through the affine stack, where weights are not constrained to be nonnegative. In the second campaign every member is admitted against the normalized-MSE network at all ten seeds: the annealed FNO at correlation 0.974–0.980 against a threshold of 0.992–0.998, the UNet at 0.943–0.961 against 0.985–1.000, the kernel at 0.859–0.879 against 0.905–0.918.

Proposition 6.7 prices what the anneal did to the FNO. In the root-mean-square metric on the validation split the member improved from 5.047% to 4.991%, by 1.1 percent, and its correlation with the normalized-MSE network rose from 0.95–0.97 to 0.974–0.980. At the first campaign's values, $t = 1.006$ and $\varrho \approx 0.96$, the exchange rate $(1 - \varrho^2)/(t - \varrho)$ is 1.7, so a 1.1-percent improvement is cancelled by a rise of about 0.019 in the correlation; the anneal produced a rise of about 0.016. The two-member optimum of the pair accordingly moved from about 4.98% to 4.97%, and the five-member prediction of Proposition 6.1 from 4.940% to 4.940%. At the second campaign's values ($t = 0.995$, $\varrho = 0.976$) the rate has risen to 2.5: the closer a member sits to the admission threshold, the more of its accuracy the stack asks it to pay for in correlation.

Part (ii) of the proposition puts the infinite-ensemble floor of an equicorrelated family at $\bar{e}\sqrt{\bar{\varrho}}$, which at these measurements is $4.922 \pm 0.056\%$ on five members and $4.910 \pm 0.057\%$ on six. Those numbers and the pipelines' 4.572% and 4.546% are not on the same footing, and the comparison has to be made carefully for the reason Remark 6.2 gives: the floor is a root-mean-square quantity and the table entries are means, so the mean lies below the floor by Jensen's inequality whatever the estimator does, and a comparison across the two would establish nothing. Converted into the floor's own metric with the measured dispersion factors, 4.572% and 4.546% correspond to about 4.87% and 4.85%, each within about one standard deviation of its floor. We therefore claim no margin below the floor. What can be said without a conversion is the qualitative point, and it is unaffected: the floor bounds convex combinations of the members, while the deployed estimator fits an affine stack per grid point and then corrects its residual with a kernel, neither of which is a convex combination; and Corollary 6.5 shows that the sign constraint is not where the difference lies, since the unconstrained global optimum coincides with the convex one at every seed. What the floor measures is how little the convex operation can buy, and the answer, at every seed and for either member set, is a few hundredths of a point.

Three further measurements are consistent with the same reading. First, the residual kernel correction, which regresses the stack's residual on the load, improves the stack by 0.024 points at $n = 19000$ whatever the kernel scale, and its tuner reaches for the smoothest kernel and the largest nugget on its grid: at this sample size the shared residual is not a function of the load in any way a Matérn RKHS on $\mathbb{R}^{41}$ can see. Second, the response to data is small. Figure S6.5 gives the learning curve at ten seeds under one protocol, the first N rows of the training pool with a fixed 1000-case validation carve and the full test block: the normalized-MSE network goes from $5.985 \pm 0.028\%$ at $N = 1000$ to $4.799 \pm 0.003\%$ at

8000 and $4.712 \pm 0.004\%$ at 19000, the kernel ridge from 6.924% to 5.444% and 5.198%. Above 8000 the network's log-log slope is -0.022 ± 0.001, fitted seed by seed, and the kernel's -0.052; the last step, 16000 to 19000, gains 0.006 ± 0.004 points, positive at nine seeds of ten, and the last two steps 0.030 ± 0.010, positive at all ten. A three-parameter fit $e_\infty + cN^{-\alpha}$ to the ten-seed means puts the network's asymptote at 4.64% and the kernel's at 4.88%, the kernel still far from its asymptotic range; the two do not agree, so the curve alone does not locate a common floor and we do not read one from it. What it fixes is the rate: at this size a doubling of the data is worth a few hundredths to the network, an order of magnitude less than the span of the published architectures. Third, and this is what ten seeds add, the error does not respond to reseeding either, and the scale of the comparison is what makes it an argument: the five published architectures in Table 2 span 0.65 points, while ten reseedings of our own pipeline span 0.034, about five percent of it. A modeling limitation would be expected to yield to capacity, diversity, data, or reseeding; within the range we could test, this error yields to none of them, which is the pattern a floor in the data would produce.

On the structural-mechanics benchmark the bracket of Corollary 6.4, with $\bar{e}^2$ the smallest diagonal entry of the measured S and the one-sided bounds read off its entries, is $[0.949 \pm 0.005,\ 1.039 \pm 0.004]$ on five members and $[0.933 \pm 0.007,\ 1.032 \pm 0.004]$ on six, about a tenth of the squared floor wide, and the normalized minimum sits inside it at every seed, at 0.980 ± 0.003 and 0.972 ± 0.005. In the root-mean-square metric $\mathcal{E}_2$ in which the corollary is stated, the best convex mixture of the measured S is at $4.940 \pm 0.057\%$ against a floor of $4.922 \pm 0.056\%$ on five members and at $4.920 \pm 0.057\%$ against $4.910 \pm 0.057\%$ on six. The bracket is evaluated against the realized ensemble risk at every seed, and its stability is the point: the realized mean relative error $\mathcal{E}_1$ of the mixture is a factor 0.938 ± 0.003 of the predicted root mean square on five members and 0.938 ± 0.003 on six, the dispersion factor of Remark 6.2, so ten independent redraws of the validation split and of every network initialization move the prediction by six hundredths of a point and the ratio by three thousandths.

The sixty-predictor pool. The measurement of Section 4.6 uses every test prediction array of the second campaign (`campaign/seedarch.py`, record `campaign/collected/dgx/seedarch.json`). The test block is split once, by a fixed permutation, into 1000 calibration and 19000 evaluation cases; second-moment and correlation matrices are formed on each half from the normalized residuals, convex optima are solved by the same sequential quadratic program as everywhere else, and seed-ensemble curves are averaged over all subsets of a given size where there are at most 45 of them and over 40 random subsets otherwise. Per architecture, the root-mean-square floor that no number of its own seeds can pass, read off the smallest within-architecture second moment, is 4.93% for the FNO, 4.96% for the normalized-MSE network, 4.97% for the refiner, 4.94% for the UNet, 4.90% for the metric-loss MLP and 5.46% for the kernel ridge; the ten-seed equal-weight averages sit within a hundredth or two of those floors, so the seed axis is exhausted at ten. The block form of Theorem 6.6 with the pool's mean second moments in place of its minima returns the uniform sixty-member mixture's value exactly, as it must, since for uniform weights the display is an identity. Fitting per-pixel affine weights over all sixty members from the 1000 calibration cases overfits, at 4.682%, where the same fit over the six ten-seed means reaches 4.556% and over the six

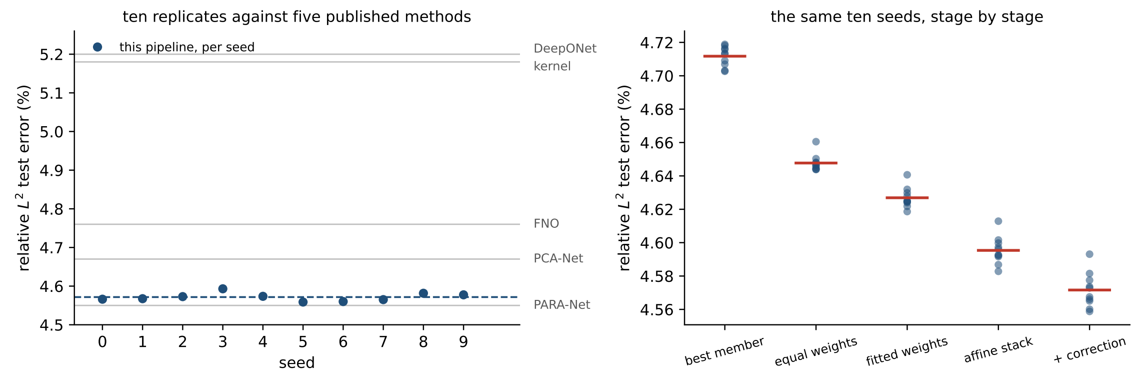

Figure S6.6: Left: the corrected pipeline's test error at each of the ten seeds (points), their mean (dashed), and the five published single numbers on the same axis (grey). The scatter of one pipeline across seeds is a small fraction of the differences the benchmark is used to adjudicate, so the published cluster cannot be attributed to run-to-run noise. Right: the same ten seeds followed down the pipeline, each stage scored on test, with the stage mean marked. Fitting per-pixel weights lowers the mean and widens the scatter; the correction lowers both.

members of seed 0 alone 4.567%. Dropping the UNet's ten seeds from the pool moves the equal-weight average from 4.611% to 4.630%.

The stages are more reproducible than the members they are built from, and the way the seed spread moves along the chain is itself informative. Equal weighting, which fits nothing, takes the members' spread down to 0.005 points on test — ordinary variance reduction. Each subsequent stage fits something, and each gives a little of that back: fitted global weights 0.006, the affine stack 0.008, the correction 0.010. The end of the chain is both the most accurate stage and the least reproducible one. Fitting is not free, and a stack that is tuned harder is not automatically a stack that reproduces better.

S7 The OCO-2 input metric, and data scaling

S7.1 What the metric buys, measured

The heuristic rate calculation of Proposition S2.5 splits a metric's advantage into a constant σ_1^s and, only under an approximate-rank gap, a rate. The seed campaign scores that split directly: both raw-input kernel rows are refit at five nested training sizes over a sixteenfold range, at ten seeds, on each of the three bands, with the sensitivity map estimated causally from the first n rows only so that no row sees data a smaller sample would not have had.

Two things come out. The approximate rank of the sensitivity map is 12 of 20 on O2 and 13–14 of 24 on the two CO_2 bands, counting the singular values that hold 99% of the squared energy. The participation ratio is a softer count and agrees on O2 (11.99 against 12) while sitting a third of a unit below the threshold count on the two CO_2 bands (13.63 against 14, 13.33 against 13); the rank ratio is between 0.54 and 0.60 on either reading,

Table S7.1: The raw-input kernel refit at nested training sizes, ten seeds per band. “Rank” counts singular values of the causally estimated sensitivity map holding 99% of the squared energy, out of the input dimension. The exponent difference is paired within seed; its standard deviation is over seeds. The last column is the isotropic-to-adapted error ratio at $n = 18000$.

Band	Rank	$-$slope iso	$-$slope adapted	difference	ratio at n=18000
O2	12/20	0.221 ± 0.007	0.215 ± 0.017	$+0.006 \pm 0.015$	1.353 ± 0.008
weak CO_2	14/24	0.086 ± 0.005	0.097 ± 0.006	-0.011 ± 0.009	1.118 ± 0.002
strong CO_2	13/24	0.203 ± 0.029	0.255 ± 0.020	-0.052 ± 0.021	1.514 ± 0.114

which is what the argument uses, and it is the regime in which the calculation would permit a rate gain and not merely a constant.

The rate gain does not appear. Table S7.1 gives the paired difference of the two fitted exponents at each band. On O2 the two exponents are indistinguishable across seeds; on the CO_2 bands the adapted exponent is steeper, but by 0.011 and 0.052 against isotropic exponents of 0.086 and 0.203, where the rank ratios above would license something several times larger. What is stable instead is the constant: at the largest training size the isotropic error exceeds the adapted error by a factor of 1.353 ± 0.008, 1.118 ± 0.002 and 1.514 ± 0.114 on the three bands, and those ratios are flat in n over the whole sweep. The reading we take is the one the measurement supports and no more: a metric is worth a constant here, the constant is between 1.1 and 1.5, and the tenfold gap to the feature kernel is not something a metric closes. Both fitted exponents are also steeper than the heuristic's own s/d, so the absolute rates are not being tested here — only the difference between them, which is what the split concerns.

At its largest size the sweep recovers the raw-kernel rows of Table 6 to the digit, seed by seed and including the selected hyperparameters. We record that as a consistency check on the two code paths and nothing more: both call the same tuning routine on the same standardized inputs, so the agreement is a re-execution of one deterministic computation rather than corroboration by an independent measurement. One protocol note: the table's matched 250-epoch budget is short of convergence for the network rows. A 750-epoch run of the same code reaches 3.82% (flat network) and 3.71% (feature head) on this band, with the combination at 3.71% reduced and 0.0174% radiance. The budget moves the levels of the trained rows together and reverses none of the orderings discussed below.

Second, the training metric behaves as a budget, and the campaign measures which part of the metric carries it. The second network in Table 6 is trained on a diagonally weighted loss over the reduced coefficients, $\|s_z \odot (\hat{z} - z)\|/\|s_z \odot z\|$. That is a surrogate for the radiance-relative error, not the error itself: the reported radiance metric maps predictions through the stored PCA reconstruction and divides by the reconstructed-radiance norm, and the surrogate omits the reconstruction. The third network is trained in the exact radiance metric, reconstruction and denominator included, and it does not win its own metric: it scores *worse* there than the surrogate-trained one, $0.0571 \pm 0.0057\%$ against $0.0442 \pm 0.0044\%$ on O2 and $0.093 \pm 0.008\%$ against $0.075 \pm 0.003\%$ on SCO2, the surrogate ahead at ten of ten seeds on both bands, with WCO2 the reverse at eight of ten ($0.064 \pm 0.015\%$ against

$0.072 \pm 0.007\%$). The kernel heads built on their features repeat the pattern. Tripling the training budget narrows the O2 gap (0.0212% against 0.0201% at 750 epochs) without reversing it, so slow optimization of the exact loss is at most part of the story. The reason the surrogate gives nothing away is the orthogonality noted above: the two losses share their numerator exactly, and differ only in the per-sample denominator, which reweights states, not coordinates. What the metric buys is coordinate weighting, and either loss buys it — the weighted networks give up the trailing coordinates the radiance barely sees and win the reported metric by a factor of three to four over the flat network, whose per-coordinate profile shows the gap directly: the kernel-flow emulator (lengthscales tuned per output) predicts the leading coefficient to 0.0012 root mean square against the flat network's 0.0054, while the flat network is three to four times more accurate on the trailing thirty coordinates. The practical rule is narrower than "train in the metric you report": train in a loss that concentrates where the reported metric concentrates. Matching the metric's weighting is what pays; matching its algebra is not.

Third, the same-class floor of the structural-mechanics study reappears here, one level down, and the campaign lets us test it rather than illustrate it. Proposition 6.1 is stated for the squared-metric risk, so the measurement is made there: for each band we form the matrix S of the ten seeded copies of the flat network on the test set, in the relative metric, and read the floor off it. Three things follow.

The identity itself holds numerically. At uniform weights $w^\top S w$ and the directly measured risk of the averaged prediction agree to 7×10^{-18}, 3×10^{-17} and 1×10^{-17} on the three bands — the proposition's content, checked against real predictions rather than assumed.

The floor is where the proposition puts it. Over all 45 seed pairs per band the residual correlations are 0.787 ± 0.010 (O2), 0.971 ± 0.001 (WCO2) and 0.961 ± 0.001 (SCO2), tight enough that the equicorrelated case of part (ii) is a fair description. It predicts a ten-member value of 4.419%, 17.997% and 9.267% against realized 4.419%, 17.997% and 9.267%, and infinite-seed floors of 4.360%, 17.970% and 9.248%. Ten seeds have already taken essentially all of what seed-ensembling has to give: the remaining distance to the infinite-seed floor is six hundredths of a point on O2 and less on the other two bands.

What the floor does not do is fall to a change of member. The comparison has to be made in one metric, since the floor is a squared-metric quantity and Table 6 reports the mean relative error $\mathbb{E}\,\ell$. Measured in the same metric as the floor, the Matérn head on learned features reaches 4.575%, 18.11% and 9.326% — better than an individual seeded network (4.915%, 18.238%, 9.433%) and worse than that network's ten-seed ensemble. The ranking is the same in the mean metric, where the seed ensemble reaches 3.916% against the head's 4.111% on O2. At this budget a different kind of member buys less than ensembling ten copies of the plain one; the order-of-magnitude result of this section is the representation ladder, which does not depend on this comparison. Ensembling within an architecture class saturates by the same second-moment law on both problems, and the two ways of spending a fixed compute budget — more seeds of one member, or one member of a better kind — are close here.

The floor is a property of the architecture class, not of the problem, and it can be probed directly. Different architectures decorrelate: on WCO2 two residual MLPs of different activation correlate at 0.84, a wide shallow network against the residual MLP at 0.40, and a random-Fourier-feature network against it at 0.08 to 0.17, all far below the 0.971 ± 0.001

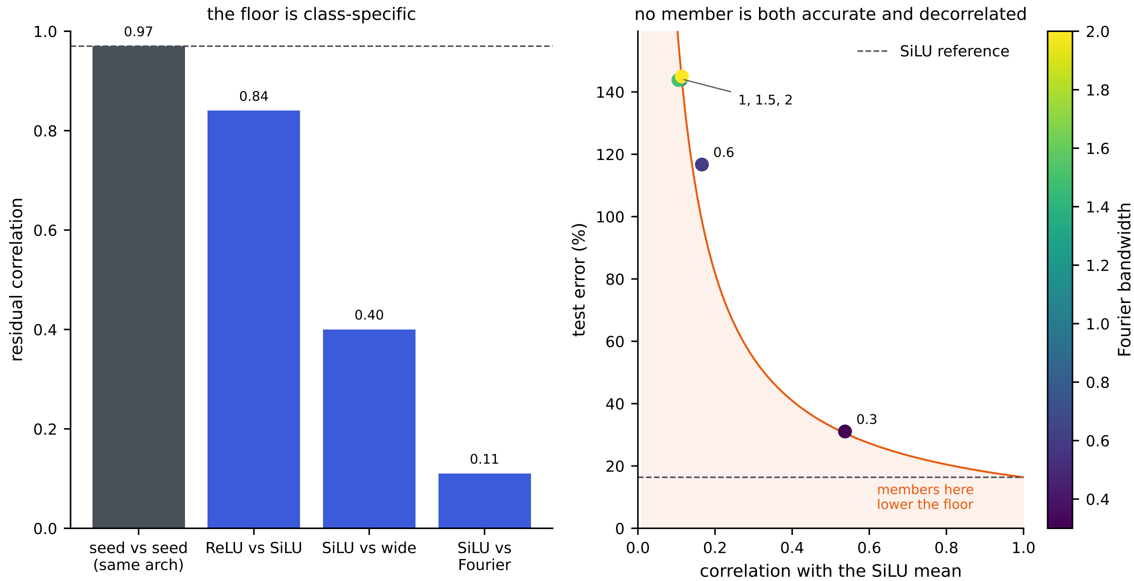

Figure S7.1: The ensembling floor on WCO2; the architecture comparison in the left panel is drawn from one seed of each, and the seed-pair correlation it shows is the campaign's mean over 45 pairs. Left: residual correlations. Two seeds of one architecture correlate near the value that sets the floor (dashed); different architectures correlate far less, so the floor is a property of the architecture class. Right: the Fourier-feature member across bandwidths. A member helps the ensemble only in the shaded region, below the e_{ref}/ϱ curve of Proposition 6.1(i); the accurate settings are too correlated and the decorrelated settings too inaccurate, so none lands there with room to spare.

that ten seeds of one architecture reach on this band. Yet the floor did not move, because its other input is accuracy: by part (i) of Proposition 6.1 a member of error e_2 helps a reference of error e_1 only when their correlation falls below e_1/e_2, and across a bandwidth sweep the Fourier network was either accurate and correlated (error 31%, correlation 0.54, just above the 0.53 threshold) or decorrelated and inaccurate (error above 140%). No architecture we tried was both accurate and decorrelated enough to lower the floor, which is why the residual MLP is the member of record: not that diversity is impossible, but that on this map no more accurate diverse member was found. Figure S7.1 shows both halves of this: the correlations that make the floor class-specific, and the accuracy-decorrelation trade-off that keeps it in place. Adding the Fourier members to the per-coordinate combination changes the reported error by less than a hundredth of a point, in either direction, on all three bands: the half-split selection gives them no weight.

Fourth, the regimes of Section 6.2 are decided by these numbers before any stack is fit. Against the flat network the raw-input kernel has $\varrho = 0.14$ and $e_n/e_k = 0.109$ (the network's 4.43% over the kernel's 40.57%): the condition of Proposition 6.1 fails, by a margin that ten seeds do not put in doubt, and the kernel is dropped — in contrast to structural mechanics where the same test keeps it. Because the radiance metric is diagonal in the reduced coordinates, the per-coordinate combination of Proposition S2.2 is optimal for either metric taken alone, at that metric's own fixed diagonal weight. It is not optimal for both at once, and the proposition does not claim to be: the two metrics normalize by different sample-dependent denominators, and both are reported as means of norms rather than as the squared risk the proposition minimizes, where Jensen gives only an upper bound. That one selector improves both is therefore something we observe on this data, not something the diagonal-weight identity implies. The selector is predeclared and shared: the same per-coordinate winners produce both columns of every band's row. It beats the kernel-flow emulator on both metrics at once, now on all three bands: O2 reaches $4.12 \pm 0.05\%$ against 16.89% reduced and $0.0294 \pm 0.0020\%$ against 0.0448% radiance, SCO2 $8.08 \pm 0.01\%$ against 16.14% and $0.0584 \pm 0.0041\%$ against 0.1147%, and WCO2 $16.28 \pm 0.06\%$ against 24.06% and $0.0507 \pm 0.0052\%$ against 0.0599%. Counted by seed, the combination beats the emulator on both metrics at ten of ten seeds on O2 and SCO2 and at nine of ten on WCO2, the single exception missing on the radiance metric by 0.0015 points.

The pool matters most on the band whose two metrics disagree. On WCO2 the flat network and its head split the metrics between them, and it is the two radiance-loss members and the heads on their features — the combination selects per coordinate from six members, the flat, weighted and exact-radiance networks and the Matérn head on each — that let one selector win both, which is what the per-coordinate argument predicts. The emulator's own predictions are not among the members, so the comparison remains against an outside baseline. A nine-of-ten margin of 0.0015 points is thin, and we report it as a band that has just crossed rather than as a comfortable win. The code releases the per-band numbers alongside this paper.

S7.2 The error is limited by data, and yields to more of it

The structural-mechanics error was flat in the sample size, which Section 4.5 reads as evidence consistent with a floor set by the benchmark data themselves, a reading whose decisive test, re-solving the benchmark on a refined mesh, has not been run; the emulator error is not flat, and the contrast is the sharpest practical difference between the two problems. The left panel of Figure S7.2 fixes the O2 task and varies only the number of training pairs, scoring each on a disjoint held-out set. The corrected surrogate's reduced-radiance error falls as a clean power law in n, fitted slope -0.68, from 74% at a few hundred pairs to 4.3% at $n = 17700$, still descending at the largest sample size we can afford and approaching the 3.8% our full pipeline reaches on all 18000 pairs (Table 6). The exact kernel on the raw state improves far more slowly, so at these sizes the network is the component that turns data into accuracy. The residual on this task is sample-limited: unlike the structural-mechanics error it recedes as pairs are added, so the largest lever on this problem is simply more of them, and pairs are cheap because the forward model is a program. The complementary case is the full-resolution $58 \to 3048$ map, where the error is instead flat in n across the few hundred retrievals we hold: that task is limited by its representation, not its sample count, and the reduction the emulator applies is what moves it.

A second benchmark carries the same reading into a data range OCO-2 cannot reach. ClimSim (Yu et al., 2023) is a climate emulation dataset of about ten million samples that maps a 124-dimensional atmospheric state to 128 physics tendencies, with published neural baselines. We run the two model families at one, two and three and a half million training

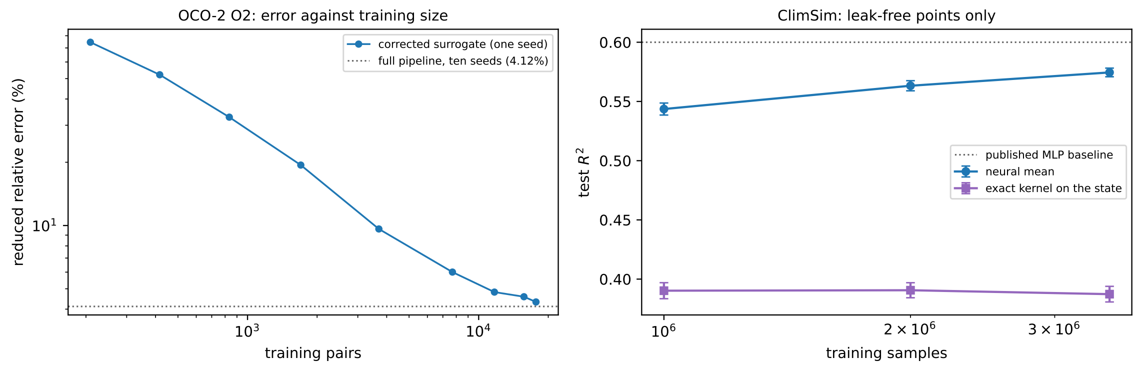

Figure S7.2: Error against training-set size, drawn by `campaign/make_scaling_fig.py`. Left: the OCO-2 O2 task, reduced error of the corrected surrogate at one seed, from the data-scaling run; it falls as a power law (fitted slope -0.68) to 4.3% at $n = 17700$, against the 4.12% the ten-seed pipeline reaches on all 18000 pairs (dotted). Right: ClimSim, test R^2 over the outputs with non-negligible variance, at one, two and three and a half million training samples under the held-out selection protocol — five seeds at one million and three at each of the larger sizes, error bars over seeds. The range begins above any plausible crossing between the two families, so none is shown; over it the kernel is flat and the neural mean is far above it and still rising toward the published baseline (dotted).

samples under the protocol used everywhere else in this paper — selection on a held-out block drawn from the training pool, the test block scored once — and the right panel of Figure S7.2 plots those points. ClimSim publishes no canonical test split, so each seed there draws its own 20000-row test block and the spreads below include that variation.

What the points establish is the high-data half of the picture, and they establish it cleanly. The exact kernel on the state is flat: 0.390 ± 0.007, 0.391 ± 0.006 and 0.387 ± 0.007 at one, two and three and a half million samples, unmoved across a factor of three and a half in data. The neural mean over the same range is both far above it and still climbing: 0.544 ± 0.005 at one million (five seeds), 0.563 ± 0.004 at two million and 0.575 ± 0.004 at three and a half million (three seeds each), closing on the published multilayer-perceptron baseline near 0.6. The separation is some twenty times the seed scatter at every size, so the ordering at these sizes is not a matter of which run one looks at, and the contrast between a data-limited mean and a saturated kernel is exactly the one the OCO-2 sweep shows on its own scale.

Where the two families cross is not something this range can locate: the sweep begins above any plausible crossing. The regimes of Section 6.2 gain a data axis from what is measured — an accurate mean is worth building once there is enough data to train it past the kernel, and at these sizes there plainly is — while the sample size at which that becomes true is an open number here, and the small-n ladder under the same protocol is the obvious next measurement.

S8 Proofs

S8.1 Proof of Proposition S1.1

By convexity of $\ell(\cdot, v)$,

$$\ell\big(\widehat{G}_S(u), G(u)\big) \;\le\; \tfrac{1}{2}\,\ell\big(\widehat{G}(u), G(u)\big) + \tfrac{1}{2}\,\ell\big(T\widehat{G}(Su), G(u)\big).$$

Since T is an involution, the equivariance $G(Su) = TG(u)$ applied at u gives $TG(Su) = T^2G(u) = G(u)$, hence

$$\ell\big(T\widehat{G}(Su), G(u)\big) = \ell\big(T\widehat{G}(Su), TG(Su)\big) = \ell\big(\widehat{G}(Su), G(Su)\big),$$

using the T-invariance of the loss. Taking expectations and using the S-invariance of μ (so that $Su \sim \mu$ when $u \sim \mu$),

$$\mathbb{E}\,\ell\big(\widehat{G}_S(u), G(u)\big) \le \tfrac{1}{2}\,\mathbb{E}\,\ell\big(\widehat{G}(u), G(u)\big) + \tfrac{1}{2}\,\mathbb{E}\,\ell\big(\widehat{G}(Su), G(Su)\big) = \mathbb{E}\,\ell\big(\widehat{G}(u), G(u)\big). \quad \square$$

S8.2 Proof of Proposition S1.2

Strict positive definiteness of the restricted kernel matrix makes I_C the (unique) interpolant of the pairs indexed by C, so $I_C(z_j) = v_j$ for $j \in C$ and the terms of (S1.1) indexed by C vanish, which is the first claim. For the second, write

$$e_2(\theta; C) \;=\; \sum_{i \in B \setminus C} g(C, i), \qquad g(C, i) := \|v_i - I_C(z_i)\|_2^2.$$

Conditionally on C, the sum has exactly $b/2$ terms, so $e_2(\theta; C) = \frac{b}{2}\,\mathbb{E}_{i \sim \mathrm{Unif}(B \setminus C)}[g(C, i)]$, and taking the expectation over C proves the identity. The gradient claim is immediate: for fixed (B, C) the map $\theta \mapsto e_2(\theta; C)$ is differentiable wherever the Cholesky factorization in I_C is (the kernel matrix stays positive definite in a neighborhood), and under a local integrable Lipschitz bound the derivative passes under the expectation, so $\mathbb{E}_{B,C}[\nabla_\theta e_2] = \nabla_\theta\, \mathbb{E}_{B,C}[e_2]$. $\square$

S8.3 Proof of Proposition S1.3

(i) Since $\sum_m w_m = 1$, $F_w(u) - G(u) = \sum_m w_m\,(f_m(u) - G(u))$, and the triangle inequality gives $\|F_w(u) - G(u)\|_2 \le \sum_m w_m \|f_m(u) - G(u)\|_2$. Dividing by $\|G(u)\|_2$ yields the pointwise claim; taking expectations gives the risk inequality, and bounding each $\ell(f_m(u), G(u))$ by B gives $\ell(F_w(u), G(u)) \le B$.

(ii) Write $\ell_u(w) = \ell(F_w(u), G(u))$. First, ℓ_u is Lipschitz on Δ for the ℓ^1 norm: for $w, w' \in \Delta$, the reverse triangle inequality and the argument of (i) give

$$|\ell_u(w) - \ell_u(w')| \;\le\; \frac{\big\|\sum_m (w_m - w'_m)(f_m(u) - G(u))\big\|_2}{\|G(u)\|_2} \;\le\; B\,\|w - w'\|_1.$$

Next, discretize the simplex. For $k \in \mathbb{N}$ let $\mathcal{G}_k = \{w \in \Delta : kw \in \mathbb{Z}^M\}$; its cardinality is the number of compositions of k into M nonnegative parts, $\binom{k+M-1}{M-1} \le (k+1)^{M-1}$. Given $w \in \Delta$, set $w'_i = \lfloor k w_i \rfloor / k$ for $i < M$ and $w'_M = 1 - \sum_{i<M} w'_i$; then $w' \in \mathcal{G}_k$ (the last

coordinate is a multiple of $1/k$ and is $\geq w_M \geq 0$ because the first $M-1$ coordinates only decreased), and

$$\|w - w'\|_1 = \sum_{i<M}(w_i - w'_i) + \big(w'_M - w_M\big) = 2\sum_{i<M}(w_i - w'_i) \;\leq\; \frac{2(M-1)}{k}.$$

By (i) the per-sample loss lies in $[0, B]$, so for each fixed w' Hoeffding's inequality gives $\mathbb{P}\big(|\widehat{R}(w') - R(w')| > t\big) \leq 2e^{-2mt^2/B^2}$, and a union bound over $\mathcal{G}_k$ shows that with probability at least $1-\delta$,

$$\sup_{w' \in \mathcal{G}_k} \big|\widehat{R}(w') - R(w')\big| \;\leq\; B\sqrt{\frac{\log\big(2|\mathcal{G}_k|/\delta\big)}{2m}}.$$

On this event, for every $w \in \Delta$, approximating by its grid point and using the Lipschitz property for both R and $\widehat{R}$,

$$\big|\widehat{R}(w) - R(w)\big| \;\leq\; B\sqrt{\frac{\log(2|\mathcal{G}_k|/\delta)}{2m}} + \frac{4B(M-1)}{k}.$$

If $w^\star$ minimizes R over Δ, the empirical minimizer satisfies $R(F_{\widehat{w}}) \leq \widehat{R}(\widehat{w}) + \sup_\Delta |\widehat{R} - R| \leq \widehat{R}(w^\star) + \sup_\Delta |\widehat{R} - R| \leq R(F_{w^\star}) + 2\sup_\Delta |\widehat{R} - R|$. Choosing $k = m(M-1)$ and using $|\mathcal{G}_k| \leq (m(M-1)+1)^{M-1}$ gives

$$R(F_{\widehat{w}}) \;\leq\; \min_{w \in \Delta} R(F_w) + \frac{8B}{m} + B\sqrt{\frac{2\big((M-1)\log(m(M-1)+1) + \log(2/\delta)\big)}{m}},$$

which is the claim. □

S8.4 Proof of Lemma S1.4

For each k the validation scores $\ell(g_k(\tilde{u}_i), G(\tilde{u}_i))$, $i \leq m'$, are i.i.d., take values in $[0, B]$, and have mean $R(g_k)$; the maps g_k are fixed relative to this sample, which is the only place the independence assumption enters. Hoeffding's inequality gives, for each k,

$$\Pr\Big(\big|\widehat{R}(g_k) - R(g_k)\big| \geq t\Big) \;\leq\; 2\exp\big(-2m't^2/B^2\big),$$

so with $t = B\sqrt{\log(2K/\delta)/(2m')}$ and a union bound over $k \leq K$, all K deviations are below t simultaneously with probability at least $1-\delta$. On that event, if k^* attains $\min_k R(g_k)$,

$$R(g_{\widehat{k}}) \;\leq\; \widehat{R}(g_{\widehat{k}}) + t \;\leq\; \widehat{R}(g_{k^*}) + t \;\leq\; R(g_{k^*}) + 2t,$$

and $2t = B\sqrt{2\log(2K/\delta)/m'}$. □

S8.5 Proof of Proposition S1.5

Fix a coordinate j and abbreviate $\varphi = \varphi_j$, $Y(u) = G(u)_j$, $D = \sqrt{M+1}$. Each entry of $\varphi(u)$ is bounded by b in absolute value (the members by assumption, the intercept entry by construction), so $\|\varphi(u)\|_2 \leq bD$, and for $\|w\|_2 \leq W$ Cauchy–Schwarz gives $|w^\top \varphi(u)| \leq WbD$

and $|w^\top\varphi(u)-Y(u)|\le b(1+WD)$. The losses $\ell_w(u)=(w^\top\varphi(u)-Y(u))^2$ therefore take values in $[0,L_\infty]$ with $L_\infty=b^2(1+WD)^2$.

Uniform deviation. Consider the one-sided suprema $Z^+=\sup_{\|w\|_2\le W}\big(\widehat{R}_j(w)-R_j(w)\big)$ and $Z^-=\sup_{\|w\|_2\le W}\big(R_j(w)-\widehat{R}_j(w)\big)$ over the validation sample of size m. Changing one sample moves either supremum by at most L_∞/m, so McDiarmid's inequality gives $Z^\pm\le\mathbb{E}Z^\pm+L_\infty\sqrt{\log(4q/\delta)/(2m)}$, each with probability at least $1-\delta/(2q)$. By symmetrization, $\mathbb{E}Z^\pm\le 2\,\mathfrak{R}_m(\mathcal{L})$, the Rademacher complexity of the loss class. The loss is the composition of $t\mapsto t^2$, restricted to $|t|\le b(1+WD)$ where it is $2b(1+WD)$-Lipschitz, with the class $\{u\mapsto w^\top\varphi(u)-Y(u)\}$; translation by the fixed function Y leaves the Rademacher average unchanged (the translated term does not involve w and has zero mean), and for the linear class over the ℓ^2 ball,

$$\mathfrak{R}_m \;\le\; \frac{W}{m}\,\mathbb{E}\Big\|\sum_{i=1}^m\sigma_i\varphi(\tilde u_i)\Big\|_2 \;\le\; \frac{W}{m}\Big(\sum_{i=1}^m\mathbb{E}\|\varphi(\tilde u_i)\|_2^2\Big)^{1/2} \;\le\; \frac{WbD}{\sqrt{m}},$$

by Jensen's inequality. Talagrand's contraction lemma then gives $\mathfrak{R}_m(\mathcal{L})\le 2b(1+WD)\cdot WbD/\sqrt{m}$, hence

$$Z^\pm \;\le\; \frac{4Wb^2D(1+WD)}{\sqrt{m}}+L_\infty\sqrt{\frac{\log(4q/\delta)}{2m}} \qquad \text{jointly with probability} \ge 1-\delta/q.$$

Empirical risk minimization. On this event, for any minimizer w^* of R_j over the ball, $R_j(\widehat{w}_j)\le\widehat{R}_j(\widehat{w}_j)+Z^-\le\widehat{R}_j(w^*)+Z^-\le R_j(w^*)+Z^++Z^-$, and Z^++Z^- is bounded by the displayed excess. A union bound over the two tails of each of the q coordinates makes all events hold simultaneously with probability at least $1-\delta$, and averaging over j preserves the (coordinate-uniform) bound. This proves (i).

For (ii), the ridge minimizer satisfies $\widehat{R}_j(\widehat{w}_j)+\lambda\|\widehat{w}_j\|_2^2\le\widehat{R}_j(w^*)+\lambda\|w^*\|_2^2$, so $\widehat{R}_j(\widehat{w}_j)\le\widehat{R}_j(w^*)+\lambda\|w^*\|_2^2\le\widehat{R}_j(w^*)+\lambda W^2$. Since by assumption $\|\widehat{w}_j\|_2\le W$, the uniform event of part (i) applies to $\widehat{w}_j$ and w^* alike, and the chain above goes through with the extra λW^2. □

S8.6 Proof of Corollary 6.4

The lower bound is Corollary 6.3. For the upper bound, evaluate the quadratic form at the uniform weights $w_m=1/M$:

$$\frac{1}{\bar e^{\,2}}\,w^\top Sw=\frac{1}{\bar e^{\,2}}\Big(\frac{1}{M^2}\sum_m S_{mm}+\frac{1}{M^2}\sum_{m\neq k}S_{mk}\Big) \;\le\; \frac{1+\delta_e}{M}+\frac{M-1}{M}\,\big(\bar\varrho+\delta_\varrho\big),$$

using the entrywise upper bounds and the counts M and $M(M-1)$ of the two sums. The right side equals $\bar\varrho+\delta_\varrho+\big(1+\delta_e-\bar\varrho-\delta_\varrho\big)/M$, and the minimum over the simplex is at most the value at any feasible point. □

S8.7 Proof of Corollary S1.6

Fix a coordinate j and a grid point γ. Substituting the affine form of the correction,

$$h_{w,\gamma}(u)_j=w_j^\top\widetilde{\varphi}_{j,\gamma}(u)+o_{j,\gamma}(u), \qquad \widetilde{\varphi}_{j,\gamma}(u)=\varphi_j(u)-\Phi_j^\top c_\gamma(u), \quad o_{j,\gamma}(u)=c_\gamma(u)^\top Y_j,$$

and neither $\widetilde{\varphi}_{j,\gamma}$ nor $o_{j,\gamma}$ depends on w or on the validation sample: c_γ is built from the training design alone. Each component of $\Phi_j^\top c_\gamma(u)$ is a c_γ-weighted sum of member values bounded by b, so its absolute value is at most $b\,\|c_\gamma(u)\|_1 \le b\kappa$, whence $\|\widetilde{\varphi}_{j,\gamma}(u)\|_2 \le bD + bD\kappa = bD(1+\kappa)$ and $|o_{j,\gamma}(u)| \le b\kappa$. For $\|w\|_2 \le W$ the prediction deviates from the target by at most $WbD(1+\kappa) + b\kappa + b \le b(1+WD)(1+\kappa) =: \tilde{c}$, so the squared losses lie in $[0, \tilde{c}^{\,2}]$ and are $2\tilde{c}$-Lipschitz in the prediction.

The argument of Proposition S1.5 now applies verbatim to the class indexed by w in the W-ball, for this fixed (j,γ): the Rademacher complexity of the affine part is at most $W\,bD(1+\kappa)/\sqrt{m}$, contraction multiplies it by $2\tilde{c}$, symmetrization doubles it, and McDiarmid at level $\delta/(2q|\Gamma|)$ adds $\tilde{c}^{\,2}\sqrt{\log(4q|\Gamma|/\delta)/(2m)}$. A union bound over the two tails of the $q|\Gamma|$ pairs makes, with probability at least $1-\delta$, each one-sided deviation $Z^{\pm}_{j,\gamma}$, defined as in the proof of Proposition S1.5, at most

$$\frac{4Wb^2D(1+WD)(1+\kappa)^2}{\sqrt{m}} + b^2(1+WD)^2(1+\kappa)^2\sqrt{\frac{\log(4q|\Gamma|/\delta)}{2m}}.$$

On that event, write $\bar{R}(\gamma) = \frac{1}{q}\sum_j R_j(h_{\widehat{w},\gamma})$ and $\widehat{\bar{R}}(\gamma)$ for its empirical version; averaging the $Z_{j,\gamma}$ over j bounds $|\widehat{\bar{R}}(\gamma) - \bar{R}(\gamma)|$ by the same quantity for every γ simultaneously, including at the data-dependent $\widehat{w}$, because the deviations are uniform over the ball. The selection chain $\bar{R}(\widehat{\gamma}) \le \widehat{\bar{R}}(\widehat{\gamma}) + Z \le \widehat{\bar{R}}(\gamma^*) + Z \le \bar{R}(\gamma^*) + 2Z$ with γ^* the aggregate oracle then gives the display, whose constant is $2Z$ in the factored form stated. When the c_γ are kernel ridge weights, Lemma S1.7 supplies the hypothesis with $\kappa = \kappa_{\mathrm{an}}$. □

S8.8 Proof of Lemma S1.7

Fix u and write $k_u = k(X,u)$, $t = n\lambda$, $A = K + tI$ and $c = A^{-1}k_u$. Because k is positive definite, the Gram matrix of the $n+1$ points $u_1,\ldots,u_n,u$ is positive semidefinite, and by the generalized Schur complement this places k_u in the range of K with $k_u^\top K^+ k_u \le k(u,u)$, K^+ the pseudo-inverse. Diagonalize $K = \sum_i \mu_i v_i v_i^\top$ and expand $k_u = \sum_i \beta_i v_i$ over the eigenvectors with $\mu_i > 0$; the Schur condition reads $\sum_i \beta_i^2/\mu_i \le k(u,u)$. Then

$$\|c\|_2^2 = \sum_i \frac{\beta_i^2}{(\mu_i+t)^2} = \sum_i \frac{\beta_i^2}{\mu_i}\cdot\frac{\mu_i}{(\mu_i+t)^2} \;\le\; \frac{1}{4t}\sum_i \frac{\beta_i^2}{\mu_i} \;\le\; \frac{k(u,u)}{4t},$$

because $(\mu+t)^2 \ge 4\mu t$ for all $\mu, t \ge 0$. Cauchy–Schwarz gives $\|c\|_1 \le \sqrt{n}\,\|c\|_2$, and combining, $\|c\|_1^2 \le n\|c\|_2^2 \le k(u,u)/(4\lambda)$.

For the sharpness take $k(x,y) = \cos(x-y)$, a positive definite kernel on the line (its native space is spanned by cos and sin), the two points $x_1 = 0$ and $x_2 = \theta$ with $\theta \in (0,\pi)$ chosen so that $1-\cos\theta = 2\lambda$, and $u = (\theta+\pi)/2$. Then $K = \left(\begin{smallmatrix} 1 & \cos\theta \\ \cos\theta & 1\end{smallmatrix}\right)$ has the eigenvector $v = (1,-1)/\sqrt{2}$ with eigenvalue $\mu = 1-\cos\theta = t$, and $k_u = (\cos u, \cos(u-\theta)) = -\sin(\theta/2)\,(1,-1)$ is proportional to v, with $\beta^2/\mu = 2\sin^2(\theta/2)/(1-\cos\theta) = 1 = k(u,u)$. Hence $c = A^{-1}k_u = k_u/(2t)$ has entries of equal magnitude, $\|c\|_2^2 = \beta^2/(2t)^2 = 1/(4t)$, and $\|c\|_1 = \sqrt{2}\,\|c\|_2 = \sqrt{n}\,\|c\|_2 = \frac{1}{2}\lambda^{-1/2}$, so every inequality in the display is an equality.

For the Matérn kernel of Section 3.3, $k(u,u) = 1$ and the smallest nugget on the grid is 10^{-6}, so $\sup_{u,\gamma}\|c_\gamma(u)\|_1 \le \frac{1}{2}\max_\gamma \lambda_\gamma^{-1/2} = 500$ irrespective of the length scale and of the design. □

S8.9 Proof of Corollary 6.5

The objective is strictly convex on the hyperplane $\{\mathbf{1}^\top w = 1\}$, so its minimizer there is the unique stationary point of the Lagrangian $w^\top S w - \nu(\mathbf{1}^\top w - 1)$, namely $w = \frac{\nu}{2} S^{-1}\mathbf{1}$ with ν fixed by the constraint: $w^\star = S^{-1}\mathbf{1}/(\mathbf{1}^\top S^{-1}\mathbf{1})$, at which $w^{\star\top} S w^\star = 1/(\mathbf{1}^\top S^{-1}\mathbf{1})$. The simplex is a subset of the hyperplane, so the convex minimum is at least this value, and by uniqueness of the minimizer the two agree exactly when $w^\star$ lies in the simplex, that is, has no negative coordinate. $\square$

S8.10 Proof of Proposition 6.1

Since $\sum_m w_m = 1$, the stack's normalized residual is $\rho_{f_w}(u) = \sum_m w_m \rho_{f_m}(u)$, hence

$$\mathbb{E}\,\ell\big(f_w(u), G(u)\big)^2 = \mathbb{E}\Big\|\sum_m w_m \rho_{f_m}(u)\Big\|_2^2 = \sum_{m,k} w_m w_k\, \mathbb{E}\,\langle \rho_{f_m}(u), \rho_{f_k}(u)\rangle = w^\top S w.$$

For (i), parameterize $w = (1-t, t)$; $q(t) = w^\top S w$ is a convex quadratic in t with $q'(0) = 2(\varrho e_1 e_2 - e_1^2)$, negative precisely when $\varrho < e_1/e_2$. In that case the unconstrained minimizer lies in $(0,1)$ and gives the stated interior value; otherwise $q'(0) \geq 0$, the minimum over $[0,1]$ is at $t = 0$ with value e_1^2, and the second member is dropped. For (ii), $S = e^2\big((1-\varrho)I + \varrho\,\mathbf{1}\mathbf{1}^\top\big)$, so $w^\top S w = e^2\big((1-\varrho)\|w\|_2^2 + \varrho\big)$ on the simplex, minimized by the uniform weights, where $\|w\|_2^2 = 1/M$. $\square$

S8.11 Proof of Corollary 6.3

Convex weights are nonnegative, so each term of $w^\top S w = \sum_m w_m^2 S_{mm} + \sum_{m\neq k} w_m w_k S_{mk}$ can be bounded below entrywise:

$$w^\top S w \;\geq\; \bar{e}^{\,2} \sum_m w_m^2 + \bar{\varrho}\,\bar{e}^{\,2} \sum_{m\neq k} w_m w_k = \bar{e}^{\,2}\Big(\|w\|_2^2 + \bar{\varrho}\,(1 - \|w\|_2^2)\Big),$$

using $\sum_{m\neq k} w_m w_k = (\sum_m w_m)^2 - \|w\|_2^2 = 1 - \|w\|_2^2$. The bracket is a convex combination of 1 and $\bar{\varrho}$, hence at least $\bar{\varrho}$. $\square$

S8.12 Proof of Theorem 6.6

Write $W_a = \sum_k w_{a,k}$ for the total weight of architecture a, so that $\sum_a W_a = 1$. Split the quadratic form into its diagonal terms, the terms between two seeds of one architecture and the terms between architectures, and bound each entry below by its hypothesis; the weights are nonnegative, so the inequalities survive the multiplication:

$$w^\top S w \;\geq\; \bar{e}^{\,2}\Big(\|w\|_2^2 + \varrho_w\big(\textstyle\sum_a W_a^2 - \|w\|_2^2\big) + \varrho_b\big(1 - \sum_a W_a^2\big)\Big),$$

using $\sum_{k\neq l} w_{a,k} w_{a,l} = W_a^2 - \sum_k w_{a,k}^2$ within an architecture and $\sum_{a\neq b} W_a W_b = 1 - \sum_a W_a^2$ across them. Rearranged, the right side is $\bar{e}^{\,2}\big(\varrho_b + (\varrho_w - \varrho_b)\sum_a W_a^2 + (1-\varrho_w)\|w\|_2^2\big)$. Both coefficients are nonnegative by hypothesis; $\sum_a W_a^2 \geq 1/M$ by Cauchy–Schwarz, with equality at $W_a = 1/M$; and $\|w\|_2^2 \geq 1/(MK)$ with equality at uniform weights, which also give $W_a = 1/M$. So the minimum of the right side over the simplex is attained at uniform weights and equals the display. When the three hypotheses hold with equality, $S = \bar{e}^{\,2}\big((1-\varrho_w)I + (\varrho_w - \varrho_b)B + \varrho_b \mathbf{1}\mathbf{1}^\top\big)$ with B the indicator of a shared architecture, the first inequality is an identity for every w, and uniform weights attain the bound. The three limits follow by letting K, then M, grow. $\square$

S8.13 Proof of Proposition 6.7

Proposition 6.1(i) gives $V = e_1^2 e_2^2(1-\varrho^2)/(e_1^2 + e_2^2 - 2\varrho e_1 e_2) = e_1^2\, t^2(1-\varrho^2)/D$ with $D = 1 + t^2 - 2\varrho t$, valid whenever the unconstrained minimizer lies in the open simplex, which is the condition $\varrho < \min(t, 1/t)$. Differentiating,

$$\frac{\partial V}{\partial t} = \frac{e_1^2\big[2t(1-\varrho^2)D - t^2(1-\varrho^2)(2t-2\varrho)\big]}{D^2} = \frac{2e_1^2\, t(1-\varrho^2)\big(D - t^2 + \varrho t\big)}{D^2} = \frac{2e_1^2\, t(1-\varrho^2)(1-\varrho t)}{D^2},$$

and

$$\frac{\partial V}{\partial \varrho} = \frac{e_1^2\big[-2\varrho t^2 D + 2t^3(1-\varrho^2)\big]}{D^2} = \frac{2e_1^2\, t^2\big(t - \varrho - \varrho t^2 + \varrho^2 t\big)}{D^2} = \frac{2e_1^2\, t^2(t-\varrho)(1-\varrho t)}{D^2}.$$

Both are positive on the interior region, since $t - \varrho > 0$ and $1 - \varrho t > 0$ there. Setting $\mathrm{d}V = (\partial V/\partial t)\,\mathrm{d}t + (\partial V/\partial \varrho)\,\mathrm{d}\varrho = 0$ and cancelling the common factor $2e_1^2 t(1-\varrho t)/D^2$ gives $(1-\varrho^2)\,\mathrm{d}t + t(t-\varrho)\,\mathrm{d}\varrho = 0$, which is the stated rate; substituting $\mathrm{d}t = -\epsilon t$ gives the fractional form. $\square$

S8.14 Proof of Proposition S2.2

The metric is diagonal, so the squared norm splits over coordinates and the j-th summand $w_j^2\,\mathbb{E}\big(\sum_m v_{mj} f_m(u)_j - G(u)_j\big)^2$ depends on the weights only through the column $v_{\cdot j}$. Minimizing the sum is therefore q independent minimizations, and the positive factor w_j^2 does not move any of the q argmins, so the optimizer is the same for every positive w. A combination with shared weights is the special case $v_{mj} = v_m$ for all j, a subset of the feasible set of each coordinate problem, which gives the domination. $\square$

S8.15 Proof of Corollary S2.3

The integrand is nonnegative, so Tonelli's theorem permits exchanging the expectation with the coordinate sum:

$$\mathbb{E}\Big[a(u)\sum_j w_j^2\big(f_v(u)_j - G(u)_j\big)^2\Big] = \sum_j w_j^2\,\mathbb{E}\Big[a(u)\big(f_v(u)_j - G(u)_j\big)^2\Big].$$

The j-th summand depends on v only through its j-th column $v_{\cdot j}$, exactly as in the proof of Proposition S2.2, so minimizing over v splits into q independent problems; the factors w_j^2 rescale the summands without moving any per-coordinate minimizer, which gives the simultaneous optimality over all diagonal w. The choice $a(u) = \|G(u)\|_2^{-2}$, $w = \mathbf{1}$ identifies the objective with $\mathbb{E}\,\|\rho_{f_v}\|_2^2$, the mean squared relative error. Finally $\mathbb{E}\|\rho\|_2 \le (\mathbb{E}\|\rho\|_2^2)^{1/2}$ is Jensen's inequality. $\square$

S8.16 Proof of Proposition S2.4

Fix a coordinate j and a member m. The validation scores $a(\tilde{u}_i)(f_m(\tilde{u}_i)_j - G(\tilde{u}_i)_j)^2$, $i \le m_v$, are i.i.d., take values in $[0, L]$ by assumption, and have mean $R_j(m)$; the members and the weight are fixed relative to the validation sample. Hoeffding's inequality gives

$$\Pr\left(\left|\widehat{R}_j(m) - R_j(m)\right| \ge t\right) \;\le\; 2\exp\bigl(-2m_v t^2/L^2\bigr)$$

for each of the Mq pairs (m, j). With $t = L\sqrt{\log(2Mq/\delta)/(2m_v)}$ and a union bound, all Mq deviations are below t simultaneously with probability at least $1-\delta$. On that event, for every j, writing $m^*(j)$ for a population minimizer,

$$R_j\bigl(\widehat{m}(j)\bigr) \le \widehat{R}_j\bigl(\widehat{m}(j)\bigr) + t \le \widehat{R}_j\bigl(m^*(j)\bigr) + t \le R_j\bigl(m^*(j)\bigr) + 2t.$$

The metric statement follows by multiplying the coordinate-wise inequality by $w_j^2 \ge 0$ and summing; the selection $\widehat{m}(\cdot)$ does not depend on w, so one event serves every diagonal metric. □

S8.17 Proof of Proposition 6.8

Let $T : \mathcal{H}_k \to \mathbb{R}^{\mathcal{X}}$ be the composition map $Tf = f \circ \varphi$. For $x \in \mathcal{X}$ the reproducing property gives $(Tf)(x) = f(\varphi(x)) = \langle f, k(\cdot, \varphi(x))\rangle_{\mathcal{H}_k}$, so evaluation of Tf at x is a bounded functional, and the image $\mathcal{H}_{k_\varphi} := T(\mathcal{H}_k)$ carries the quotient norm $\|g\|_{\mathcal{H}_{k_\varphi}} = \min\{\|f\|_{\mathcal{H}_k} : Tf = g\}$, the minimum over the affine subspace $T^{-1}(g)$ (nonempty exactly when g is a pullback). This normed space is an RKHS: its reproducing kernel is $k_\varphi(x, x') = \langle k(\cdot, \varphi(x)), k(\cdot, \varphi(x'))\rangle_{\mathcal{H}_k} = k(\varphi(x), \varphi(x'))$, since the minimum-norm representer of evaluation at x is $k(\cdot, \varphi(x))$. Taking $f = h$ in the minimum gives $\|h \circ \varphi\|_{\mathcal{H}_{k_\varphi}} \le \|h\|_{\mathcal{H}_k}$, and substituting this bound into Theorem 6.9 applied with the kernel k_φ replaces $\|G\|$ by $\|h\|_{\mathcal{H}_k}$. □

S8.18 Heuristic derivation of Proposition S2.5

The proposition is labelled informal, and this appendix records the calculation behind it at the level of the idealization stated with it — projected image marginal bounded above and below, exact rank at the working resolution, and the interpolation limit standing in for the validated nugget — as a sketch, without tracking constants; no lower bound is claimed. The argument has five steps: a pointwise error bound whose design dependence is the power function; a fill-distance bound for random designs; an upper bound on the warped target's native norm; the two substitutions; and the reduction from the validated nugget to the interpolation scale. Throughout, k is a Matérn kernel of smoothness s whose native space on a bounded Lipschitz domain $\Omega \subset \mathbb{R}^{d'}$ is norm-equivalent to $H^{s+d'/2}(\Omega)$ (Wendland, 2004, Ch. 10); we follow the convention of the statement and write s for the resulting approximation order.

Step 1: pointwise bound through the power function. For any target F with components in the native space and the interpolant $\widehat{F}_0$ on the design X, the argument that proves Theorem 6.9 (with $\lambda = 0$ and $m = 0$) gives, for every u,

$$\|F(u) - \widehat{F}_0(u)\|_2 \;\le\; \|F\|_K\, P_0(u),$$

and the classical zeros lemma for functions with many zeros on a dense set converts the power function into the fill distance $h_X = \sup_{u\in\Omega}\min_i \|u-u_i\|$: $\sup_u P_0(u) \le C\, h_X^s$ for h_X below a domain constant (Wendland, 2004, Ch. 11). Hence $\sup_u \|F(u)-\widehat{F}_0(u)\|_2 \le C\, h_X^s\, \|F\|_K$.

Step 2: fill distance of an i.i.d. design. Let $u_1,\dots,u_n$ be i.i.d. from a density bounded below by $p_- > 0$ on Ω. Fix $\delta > 0$ and a $\delta/2$-net of Ω of cardinality $N(\delta) \le C_\Omega\, \delta^{-d'}$. A net ball of radius $\delta/2$ has μ-mass at least $c\,p_-\delta^{d'}$, so the probability that some net ball receives no sample is at most $C_\Omega\, \delta^{-d'}(1-c\,p_-\delta^{d'})^n \le C_\Omega\, \delta^{-d'} e^{-c\,p_-\, n\delta^{d'}}$. Choosing $\delta^{d'} = (2/(c\,p_-))\,(\log n)/n$ makes this at most $C'\, n^{-1}$, and on the complement every point of Ω lies within δ of a sample: with probability at least $1 - C'/n$,

$$h_X \;\le\; C''\Big(\frac{\log n}{n}\Big)^{1/d'}.$$

This is the high-probability statement in the proposition; the logarithm is absorbed into the logarithmic factors there.

Step 3: the warped target's norm, from above. Let $F = g(A\,\cdot)$ with $\|g\|_{H^s} = 1$ (native norm of the isotropic kernel in the image variables). For integer m, the chain rule gives, for any multi-index α with $|\alpha| = m$, $D^\alpha\{g(Au)\} = \sum_{|\beta|=m} c_{\alpha\beta}(A)\,(D^\beta g)(Au)$ with $\sum_\beta |c_{\alpha\beta}(A)| \le \|A\|^m = \sigma_1^m$, so by the change of variables $y = Au$ (Jacobian $|\det A|$),

$$\|D^\alpha\{g(A\cdot)\}\|_{L^2(\mathcal{U})} \;\le\; \sigma_1^m\,|\det A|^{-1/2}\,\|g\|_{H^m(A\mathcal{U})}.$$

Summing over $|\alpha| \le m$ bounds $\|g(A\cdot)\|_{H^m}$ by $C\,\sigma_1^m|\det A|^{-1/2}\|g\|_{H^m}$ for integer m, and the fractional case follows by interpolation between consecutive integers (Adams and Fournier, 2003, Ch. 7). Only this upper bound is used; the $|\det A|^{-1/2}$ and the domain constants are among the singular-value constants absorbed into the statement, leaving the displayed σ_1^s.

Step 4: the two designs. For the isotropic kernel on the raw input the domain has dimension $d' = d$, so Steps 1–3 give the first display: $C\, h_X^s\, \sigma_1^s \lesssim \sigma_1^s n^{-s/d}$ up to logarithms. For the adapted kernel $k_A(u,u') = k(Au, Au')$, interpolating F at $\{u_i\}$ is interpolating g at the image design $\{Au_i\}$, and $\|F\|_{K_A} = \|g\|_K \le 1$ by the pullback identity (Proposition 6.8). The image measure has a density bounded above and below on its support by the idealization of the statement. Split $A = A_r + A_\perp$ along the leading-r right singular subspace. The projected points $\Pi_r A u_i$ are i.i.d. on an r-dimensional bounded set, so Step 2 (with $d' = r$) puts every projected image point within $C(\log n/n)^{1/r}$ of a projected sample; and for every u in the domain, $\|Au - \Pi_r A u\| \le \sigma_{r+1}\operatorname{diam}(\mathcal{U})$. The triangle inequality then bounds the image fill distance by $C(\log n/n)^{1/r} + 2\sigma_{r+1}\operatorname{diam}(\mathcal{U})$, and the approximate-rank hypothesis $\sigma_{r+1} \lesssim n^{-1/r}$ makes the second term of the same order as the first. Steps 1 and 3 with $d' = r$ and $\|g\|_K \le 1$ give the second display. If instead σ_{r+1} is not below $n^{-1/r}$, the trailing directions are resolved by the design, the image fill distance is that of a d-dimensional set, and the two rates coincide; only the constant σ_1^s separates the bounds, as claimed.

Step 5: from the interpolant to the validated ridge estimator. The pipeline fits with a nugget from a grid containing values down to $10^{-8}n$ and selects on validation. For any fixed λ the pathwise bound of Theorem 6.9 controls the ridge estimator by $\|F\|_K \sup_u P_\lambda(u)$, and a two-line spectral computation compares the regularized power function with the interpolation one: with $k_u = k(X,u)$ and $A_\lambda = K + n\lambda I$,

$$P_\lambda(u)^2 - P_0(u)^2 = k_u^\top\big(K^{-1} - A_\lambda^{-1}\big)k_u = n\lambda\, k_u^\top K^{-1} A_\lambda^{-1} k_u \;\le\; \frac{n\lambda\,\|k_u\|_2^2}{\lambda_{\min}(K)\,\big(\lambda_{\min}(K) + n\lambda\big)},$$

which vanishes as $\lambda \to 0$. Validation selection over the grid can only improve on its smallest-nugget arm up to the selection cost priced by Lemma S1.4, which is of lower order than the rates displayed. Both displays are upper bounds under the stated idealization, and no matching lower bound is claimed. □

S8.19 Proof of Theorem 6.9

Fix u and write $k_u = k(X,u) \in \mathbb{R}^n$, $A = K + n\lambda I$, $w = A^{-1}k_u$. For each component j, membership $r_j \in \mathcal{H}_k$ and the reproducing property give

$$r_j(u) - \widehat{r}_{\lambda,j}(u) = \Big\langle r_j,\ k(u,\cdot) - \sum_{i=1}^{n} w_i\, k(u_i,\cdot)\Big\rangle_{\mathcal{H}_k},$$

because $\widehat{r}_{\lambda,j}(u) = k_u^\top A^{-1} r_j(X) = \sum_i w_i\, r_j(u_i)$ and $r_j(u_i) = \langle r_j, k(u_i,\cdot)\rangle$. Cauchy–Schwarz yields

$$\big|r_j(u) - \widehat{r}_{\lambda,j}(u)\big| \ \le\ \|r_j\|_{\mathcal{H}_k} \Big\|k(u,\cdot) - \sum_i w_i k(u_i,\cdot)\Big\|_{\mathcal{H}_k},$$

and the second factor is independent of j; call it $\rho(u)$. Expanding,

$$\rho(u)^2 = k(u,u) - 2\, w^\top k_u + w^\top K w = k(u,u) - k_u^\top A^{-1}\big(2A - K\big)A^{-1}k_u.$$

Since $2A - K = A + n\lambda I$,

$$\rho(u)^2 = k(u,u) - k_u^\top A^{-1}k_u - n\lambda\, \|A^{-1}k_u\|_2^2 = \widetilde{P}_\lambda(u)^2 \ \le\ P_\lambda(u)^2.$$

Summing the squared componentwise bounds,

$$\big\|r(u) - \widehat{r}_\lambda(u)\big\|_2^2 = \sum_j \big(r_j(u) - \widehat{r}_{\lambda,j}(u)\big)^2 \le \rho(u)^2 \sum_j \|r_j\|_{\mathcal{H}_k}^2 = \widetilde{P}_\lambda(u)^2\, \|r\|_K^2.$$

For the sharpness, write $g_u = k(u,\cdot) - \sum_i w_i k(u_i,\cdot)$, so that $\|g_u\|_{\mathcal{H}_k} = \rho(u) = \widetilde{P}_\lambda(u)$. If $\widetilde{P}_\lambda(u) = 0$ both sides of the bound vanish for every r. Otherwise take $r_1 = (\|G - m\|_K/\rho(u))\, g_u$ and $r_j = 0$ for $j \ge 2$: this residual has norm $\|G - m\|_K$, and since the estimator is built from its own training values $r_1(X) = g_u(X)$, the first display gives $r_1(u) - \widehat{r}_{\lambda,1}(u) = \langle r_1, g_u\rangle_{\mathcal{H}_k} = \|G - m\|_K\, \rho(u)$, equality in Cauchy–Schwarz. □

Note that the argument does not use interpolation: it holds for every $\lambda \ge 0$, with the (slightly sharper) factor $\widetilde{P}_\lambda$ showing that smoothing can only tighten this particular bound relative to the posterior standard deviation P_λ that we report.

S8.20 Proof of Theorem 6.10

Write $k_u = k(X,u)$ and let $g_u = k(u,\cdot) - \sum_i (K^{-1}k_u)_i k(u_i,\cdot)$ be the function of the previous proof at $\lambda = 0$. It vanishes on the design, since $g_u(u_j) = k(u,u_j) - k_u^\top K^{-1}K e_j = 0$, and $g_u(u) = \|g_u\|_{\mathcal{H}_k}^2 = k(u,u) - k_u^\top K^{-1}k_u = P_0(u)^2$. If $P_0(u) = 0$ the lower bound is trivial. Otherwise set $r^\pm = \pm(\rho/P_0(u))\, g_u\, e_1$, two residual operators of norm ρ whose training values are both zero, so that any Ψ returns the same vector $\psi = \Psi(0)$ for both; their values at u are $\pm\rho P_0(u) e_1$, hence $\max_\pm \|r^\pm(u) - \psi\|_2 \ge \max_\pm |\pm\rho P_0(u) - \psi_1| \ge \rho P_0(u)$. This is the lower bound. Theorem 6.9 with $\lambda = 0$ bounds the interpolant's error by $\|r\|_K P_0(u) \le \rho P_0(u)$ on the whole ball, so the interpolant attains it, and for $\lambda > 0$ the same theorem together with its sharpness case gives the ridge correction's worst case as exactly $\rho\, \widetilde{P}_\lambda(u)$.

For the identity, put $t = n\lambda$ and $A = K + tI$, which commutes with K. From the proof of Theorem 6.9, $\widetilde{P}_\lambda(u)^2 = k(u,u) - k_u^\top A^{-1}(2A - K)A^{-1}k_u$, while $P_0(u)^2 = k(u,u) - k_u^\top K^{-1}k_u$, so

$$\widetilde{P}_\lambda(u)^2 - P_0(u)^2 = k_u^\top\big(K^{-1} - 2A^{-1} + A^{-1}KA^{-1}\big)k_u = k_u^\top A^{-2}K^{-1}\big(A^2 - 2AK + K^2\big)k_u = k_u^\top A^{-2}K^{-1}(A-K)^2 k_u,$$

and $A - K = tI$. The right-hand side is nonnegative because $A^{-2}K^{-1}$ is positive definite, which gives $P_0 \le \widetilde{P}_\lambda$; the upper inequality $\widetilde{P}_\lambda \le P_\lambda$ was shown above. Expanding k_u in the eigenbasis of K as in the proof of Lemma S1.7, the difference equals $\sum_i (\beta_i^2/\mu_i)\,(t/(\mu_i + t))^2$, each term of which is bounded by β_i^2/μ_i and tends to zero with t, so the difference vanishes as $\lambda \to 0$ by dominated convergence. □

S8.21 Proof of Proposition S3.1

Write $s_i = \|e(\tilde{u}_i)\|_2 / P_\lambda(\tilde{u}_i)$ for the validation scores and $s^\star = \|e(u^\star)\|_2 / P_\lambda(u^\star)$ for the test score. Because m, $\widehat{r}_\lambda$, X and P_λ are fixed and the inputs $\tilde{u}_1, \dots, \tilde{u}_m, u^\star$ are exchangeable and independent of them, the scores $s_1, \dots, s_m, s^\star$ are exchangeable. The event $G(u^\star) \in C(u^\star)$ is exactly $\{s^\star \le q\}$, where $q = s_{(\lceil (1-\alpha)(m+1) \rceil)}$ is the $k := \lceil (1-\alpha)(m+1) \rceil$-th order statistic of the validation scores.

Consider the augmented sample $s_1, \dots, s_m, s^\star$ of size $m+1$ and let $\mathrm{rank}(s^\star)$ be its rank (ties broken uniformly at random, which only helps). By exchangeability the rank is uniform on $\{1, \dots, m+1\}$, so $\mathbb{P}(\mathrm{rank}(s^\star) \le k) = k/(m+1)$. If $s^\star$ is among the k smallest of the $m+1$ values then at most $k-1$ of the s_i are below it, so $s^\star \le s_{(k)} = q$; conversely $s^\star \le q$ forces $\mathrm{rank}(s^\star) \le k$ except possibly through ties. Hence

$$\mathbb{P}(s^\star \le q) \;\ge\; \mathbb{P}(\mathrm{rank}(s^\star) \le k) = \frac{k}{m+1} = \frac{\lceil (1-\alpha)(m+1) \rceil}{m+1} \;\ge\; 1 - \alpha,$$

which is the lower bound. When the scores are almost surely distinct there are no ties, $\{s^\star \le q\} = \{\mathrm{rank}(s^\star) \le k\}$ exactly, and $k/(m+1) < ((1-\alpha)(m+1) + 1)/(m+1) = 1 - \alpha + 1/(m+1)$, giving the upper bound. □

S8.22 Proof of Corollary S5.2

The nonzero eigenvalues of $K/n = XX^\top/n$ coincide with those of the sample covariance matrix $S = X^\top X/n \in \mathbb{R}^{p \times p}$, and zero eigenvalues contribute nothing to $d_{\mathrm{eff}}(\lambda) = \sum_i \lambda_i(K)/(\lambda_i(K) + n\lambda)$, so

$$\frac{d_{\mathrm{eff}}(\lambda)}{n} = \frac{p}{n} \cdot \frac{1}{p} \sum_{j=1}^{p} \frac{\lambda_j(S)}{\lambda_j(S) + \lambda} = \frac{p}{n}\Big(1 - \lambda\, \widehat{m}_S(-\lambda)\Big),$$

by the computation of Lemma S5.1 applied to S, with $\widehat{m}_S$ the Stieltjes transform of the empirical spectral distribution of S. By the Marčenko–Pastur theorem (Marčenko and Pastur, 1967), this distribution converges weakly, almost surely, to the law with ratio γ and scale σ^2, and since $x \mapsto (x+\lambda)^{-1}$ is bounded and continuous on $[0, \infty)$, $\widehat{m}_S(-\lambda) \to m(-\lambda)$.

It remains to evaluate $m(-\lambda)$. The Stieltjes transform of the Marčenko–Pastur law satisfies the self-consistent equation

$$m(z) = \frac{1}{\sigma^2\big(1-\gamma-\gamma z\, m(z)\big) - z},$$

i.e. $\gamma\sigma^2 z\, m^2 + \big(z - \sigma^2(1-\gamma)\big)m + 1 = 0$. At $z = -\lambda$ the discriminant is $\big(\lambda + \sigma^2(1-\gamma)\big)^2 + 4\gamma\sigma^2\lambda > 0$, and of the two real roots

$$m(-\lambda) = \frac{\sigma^2(1-\gamma) + \lambda \pm \sqrt{\big(\lambda + \sigma^2(1-\gamma)\big)^2 + 4\gamma\sigma^2\lambda}}{-2\gamma\sigma^2\lambda}$$

only the one with the minus sign in the numerator is positive, as $m(-\lambda) = \int (x+\lambda)^{-1} dF(x) > 0$ requires; rearranging gives the stated expression. $\square$

S8.23 Proof of Lemma S5.1

Diagonalize K with eigenvalues λ_i. Then

$$\operatorname{tr}\big(K(K+n\lambda I)^{-1}\big) = \sum_{i=1}^{n} \frac{\lambda_i}{\lambda_i + n\lambda} = \sum_{i=1}^{n} \Big(1 - \frac{n\lambda}{\lambda_i + n\lambda}\Big) = n - \lambda \sum_{i=1}^{n} \frac{1}{\lambda_i/n + \lambda} = n\big(1 - \lambda\, \widehat{m}(-\lambda)\big),$$

using $\widehat{m}(-\lambda) = \frac{1}{n}\sum_i (\lambda_i/n + \lambda)^{-1}$. The limit statement follows from the weak convergence of the empirical spectral distribution together with the boundedness and continuity of $x \mapsto (x+\lambda)^{-1}$ on $[0,\infty)$ for fixed $\lambda > 0$. $\square$

S9 Implementation details

The pipeline was developed on a single laptop (one NVIDIA RTX PRO 2000, 8 GB; 16 CPU cores, 64 GB RAM); the structural-mechanics numbers were then recomputed from scratch on CPU-only cloud instances under the seed protocol below, in single precision for network training and double precision for all kernel solves and error computations, and it is those recomputed values that the tables report; the second campaign of Section 4.1 ran on one 40-core CPU host (Intel Xeon E5-2698 v4) under the same protocol. The run records carry no device field, but each network trainer prints the device it selected when it starts, and the ten structural-mechanics pipeline logs hold that line for every launch, restarts included: 54 launches, all `cpu`. The kernel stages are NumPy and SciPy and run on the CPU by construction. The dataset is the distributed `StructuralMechanics` file pair (40000 samples); inputs are reduced to $\mathbb{R}^{41}$ after verifying the broadcast structure exactly. Splits: the training block is samples 1–20000, the test set is samples 20001–40000. High-data protocol: a fixed permutation of the training block reserves 1000 samples for validation, 19000 for fitting. Low-data protocol: the first 1250 samples of the training block, of which the last 250 form the validation split. The test set is never touched during development; each configuration is evaluated on it once, after selection on validation.

FNO. Width 64, 14 modes per dimension, 4 spectral layers with pointwise linear skips and GELU, lift from 3 channels (broadcast load, two coordinates), projection $64 \to 128 \to 1$. AdamW, learning rate 1.5×10^{-3} (2×10^{-3} with batch 256), weight decay 10^{-6}, cosine schedule to 10^{-6}, 200 epochs, batch 256. 6.45M parameters.

Transformer (implemented, not in the reported ensemble). Encoder: 41 tokens (load value and 10 sine/cosine pairs of the coordinate), 5 pre-norm blocks, dimension 192, 4 heads, MLP ratio 4. Decoder: grid queries from Fourier features of both coordinates, two cross-attention blocks, pointwise head $192 \to 192 \to 1$; 3.16M parameters. The cross-attention over 1681 grid queries is the most compute-intensive of our means, and the hardware available for this study (a single laptop GPU that proved unstable under sustained load) did not let us train it to the level of the other members; we therefore exclude it from the reported ensemble and note it as the natural fourth architecture family for a follow-up on stable hardware.

UNet. Three convolutional scales ($41 \to 21 \to 11$ by average pooling, ceil mode) and a bottleneck at 6×6; two 3×3 convolutions with GroupNorm(8) and SiLU per block; widths 48/96/192/384; bilinear upsampling with skip concatenation; 1×1 output head. Input channels as for the FNO. AdamW, learning rate 1.5×10^{-3}, weight decay 10^{-5}, cosine schedule, 200 epochs, batch 256. 4.38M parameters.

MSE variant. The residual MLP trained with mean squared error on standardized targets in place of the metric loss, otherwise identical recipe but for the schedule: the campaign ran it for 120 epochs; it enters the ensemble as a loss-diversity member and is also the strongest plain MLP we obtain.

MLP. Input layer $41 \to 1024$, three residual SiLU blocks of width 1024, output $1024 \to 1681$. AdamW, learning rate 10^{-3}, weight decay 10^{-5}, cosine schedule, 400 epochs, batch 256. 4.91M parameters. A wider variant (width 1536, five residual blocks, 14.5M parameters) is trained under the same recipe.

Refiner. The same residual trunk with input layer $(41 + 1681) \to 1024$: the load is concatenated with the kernel method's stress prediction for that load. Training uses four-fold out-of-fold kernel predictions for the input channel and full-data kernel predictions at evaluation; otherwise identical to the MLP but run for 100 epochs (the trainer's own default of 300 is not what the campaign used). 6.64M parameters. Reflection augmentation flips the load and permutes both the kernel channel and the target by the row-reversal index.

Common training details. Loss: mean over the batch of $\|\hat{v} - v\|_2/\|v\|_2$ in original units (predictions denormalized inside the loss; targets standardized by the per-pixel training mean and global standard deviation), except for the MSE variant above. Reflection augmentation with probability 1/2; reflection averaging at evaluation. Model selection: validation error computed every 10 epochs, best checkpoint kept. The ensemble's diversity comes from the architectures and losses rather than from reseeding; the correlation analysis of Section 4.5 indicates same-architecture reseeds would be more correlated still.

Seed protocol. One seed value enters every stochastic component at once: the permutation that draws the validation split from the training pool, each member's initialization and batch order, the half-split of the stacking comparison, the subsamples used to tune the kernel stages, and the conformal calibration draw; the pool/test boundary itself never moves. ClimSim is the one exception, and deliberately so: that benchmark fixes no test split of its own, so the seed there also draws the 20000-row test block out of the full set (`campaign/climsim_seeded.py`). Its seed spreads therefore include test-sampling variation, which the structural-mechanics and OCO-2 spreads exclude by construction; that makes them the more conservative of the two conventions where ClimSim is concerned, and it is why the ClimSim seed scatter is quoted as a check on the crossing rather than as a precision claim. Tables report the mean and standard deviation across seeds. What the repository carries is the per-seed result record for every stage of every run — the JSON files the tables and this appendix are computed from, together with the scripts that compute them. The network checkpoints and the full prediction arrays those records were derived from are too large for the repository; what is released in their place, for the structural-mechanics chain, is the residual of every member at every seed, in the form described under "Released residuals" below. A reader can therefore recheck every reported number against the records, rescore a new combination rule on the stored residuals, and rerun the pipeline from the code. Where a statement below turns on an array rather than on a record, it is marked.

What the two campaigns completed. The first structural-mechanics chain was enqueued at ten seeds on four CPU instances. The kernel ridge, the metric-loss MLP, the normalized-MSE MLP and the refiner completed at all ten, forty trained artifacts in total. The campaign's 36-hour per-task limit then landed inside the FNO stage at every seed. The FNO has a best-validation checkpoint at all ten seeds, written by the trainer each time validation improved, and those checkpoints are finalized by running the tail of the trainer on them (evaluate, then save the named artifact), so the FNO of that campaign is a ten-seed member and nothing about it is retrained. Its training curves bound what the early stop cost: the best validation epoch falls between 10 and 70 of the scheduled 200, and at every seed validation was rising again by the last epoch reached, in several cases sharply (seed 3, 0.0489 at its best against 0.0550 at epoch 150; seed 5, 0.0483 against 0.0557), so the member is the early-stopped optimum a full run would also have kept, short of the anneal. The UNet, scheduled after the FNO, did not run inside that limit. With the FNO finalized, all ten seeds carry the five-member stack, the residual correction, the second-moment measurement and the conformal calibration; the four-member versions of each are retained and reported so that the FNO's contribution is a measurement rather than an assumption. One instance was killed by the operating system while forming the 19000×19000 Gram matrix through full-size temporaries (the released code fills it in row blocks instead, Section S6.4), and four seeds hit the task limit during stack stages and were completed on a larger instance afterwards.

The second chain ran the same ten seeds on one 40-core CPU host. It trained the kernel ridge and the three MLP-family members again under the same seeds — seeded CPU training is deterministic, and their test errors agree with the first campaign's to within 6×10^{-8} in the relative-error unit for the two MLPs and the kernel ridge and 6×10^{-6} for the refiner — and then the FNO and the UNet under a 100-epoch cosine schedule, each to the end of it: nine of the ten FNO runs and all ten UNet runs keep their final checkpoint, the tenth FNO its epoch-79 one. The schedule is normalized to its own budget, so it anneals fully to 10^{-6}; it is a complete shorter schedule, not the first campaign's schedule cut short. The per-pixel stack, the residual correction, the global convex stack, the second-moment measurement and the conformal calibration were then run on the six members at every seed and, for the paired comparisons of Section 4.3, on the same five without the UNet; in that five-member run the half-split rule declined the per-pixel weights at one seed, where the deployed pipeline is the corrected global convex stack, as the rule prescribes. The member cost is recorded in Table S9.2.

The OCO-2 and ClimSim sweeps of Section S7.2 report their own seed counts, which are ten per band and five, three and three by training size respectively. The low-data chain completed at ten seeds. The OCO-2 head pipeline is replicated at ten seeds on each of the three bands — thirty band-seeds at a matched 250-epoch budget, with one further run at 750 epochs to separate budget effects from ordering effects. Each band-seed writes its member predictions and per-sample errors, which is what makes the combination rules, the floors and the scoreboard of Section 5 recomputable without retraining; those arrays stay with the run and are not part of the distributed release. Seven of the thirty band-seeds were run a second time on different hardware, which serves as a reproducibility check: six of the ten rows reproduce to the printed precision across the two runs, including the kernel rows, the flat-metric network, its feature head and the combination that the paper reports, while the three weighted-loss members and their heads move by up to 0.39 points. Seeding fixes the initialization and the data order but not every reduction the training performs, so a seeded rerun of those members should be expected to land nearby rather than exactly; the tables' standard deviations, which are over seeds, are larger than this in every case.

Reconciling the stored arrays. Every member's saved prediction arrays are checked against the metric recorded in its own training record before any ensemble is formed: the array is rescored under the project's metric and the two must agree. The check matters because each seed has its own validation partition, and scoring one seed's predictions against another's split produces errors near 28% that are otherwise easy to mistake for a training failure — and, worse, produces no discrepancy at all on the one seed whose split happens to be the reference. Reconciliation is run at every seed and every member. On the test block, which no seed moves, all forty pairs agree with their recorded reflection-averaged error to between 10^{-14} and 6×10^{-8}, the precision at which the arrays are stored.

Released residuals. The residual and selection analyses of Sections 4.5 and 4.3 consume the member residuals $r_m = f_m - y$ rather than the predictions, and the residuals are what we release. For each of the ten seeds one compressed NumPy archive, `sm_residuals_s`k`.npz`, holds the test-block residual of each of the five members and of the reported pipeline (per-pixel stack plus kernel correction) as a 20000×1681 array, the validation-split residual of each member as a 1000×1681 array, the target norms $\|y\|$ of every sample on both splits, and the index arrays of both splits; the residuals are stored in IEEE half precision (`float16`), the norms and indices at full width. The archives were written from the reconciled arrays above by `campaign/export_residuals.py`, which also recomputes every member's validation and test error from the half-precision residual and compares it with the stage record the tables are built from: over all 110 arrays the two agree to within 5.4×10^{-8} in the relative-error unit, and the largest residual in any array is 192 against target norms of 3.7×10^{3} to 1.7×10^{4}, so the rounding is far below the seed-to-seed spread of any quantity in the paper. At 384 MB per seed, 3.84 GB in all, the archives exceed what the code repository can carry; they are to be deposited with the published version, and the repository keeps the manifest (`runs/residuals_manifest.json`: per array the seed, the split seed, member, split, shape, dtype, the SHA-256 of its bytes, the largest residual and the three errors just described; per archive its size and SHA-256), so a deposited file can be checked byte for byte against the version the paper was computed from. From these files alone, without the dataset, a reader can recompute every member error of Table 4 as the mean of $\|r_m\|/\|y\|$ over samples; the second-moment matrix S, the simplex weights, the predicted and realized risks and the dispersion factor of Proposition 6.1 (`campaign/secmom_seeded5.py` does nothing else with the arrays than form $r_m/\|y\|$ on the validation split and evaluate $\|\sum_m w_m r_m\|/\|y\|$ on test, which is exact because the weights sum to one); the pairwise residual correlations and the equicorrelated floor; the equal-weight and fitted ensembles, the spatial error maps and the hard-sample comparison of Section 4.5; and the raw conformal scores $\|r\|$ of the pipeline, calibrated on the same seed-drawn subset of the test block. The per-pixel affine refit and the kernel correction need the predictions themselves, $f_m = y + r_m$, and the kernel's posterior scale needs the Gram matrix; both follow from the distributed dataset and the released code. The network checkpoints, the training-split predictions and the OCO-2 and ClimSim arrays remain with the runs, as stated where they arise.

Compute record. The campaign recorded wall clock but not memory, and Table S9.1 reports what it recorded; nothing in it is a new measurement. Each member's trainer writes its own elapsed minutes and the epoch of the checkpoint it kept into its run record, prints a cumulative timer every ten epochs, and prints the device it selected; the kernel stage prints the elapsed seconds of every grid cell, one Cholesky factorization and solve of the 19000×19000 system followed by the validation prediction, and the out-of-fold stage the seconds per fold at $n = 14250$. Every task ran six threads, five tasks to an instance, so the figures are wall clock under the campaign's own load rather than the cost of an isolated job, and the spread across seeds is mostly the spread across hosts: the four seeds the larger instance lists as its own in the campaign's task records (2, 4, 8 and 9) are separated from the other six by the kernel's per-cell time without overlap, 20–35 s against 42–113 s, and the table keeps the two classes apart. In this campaign the FNO has no completed record, and its rate is read from the ten-epoch timer prints: the uninterrupted intervals run at 467–625 s per epoch on the standard instances and 348–410 s on the larger one, and the remaining intervals, which reach 10^4 s per epoch, record contention and suspension on the shared hosts rather than the method, which is why the table gives the median of all intervals alongside the minimum. The checkpoints that were kept sit at epochs 10 to 70 and were reached after 1.3–8.0 h of training at the five seeds whose timer ran uninterrupted to that epoch, and after 9.0–16.9 h at the other five, whose timers include such periods. The inference time for the test block was not recorded for any member (the prediction logs carry the score only), and no stage recorded peak memory; the out-of-memory marker at seed 4 is the Gram-matrix failure described above. The harvest, with the per-seed values behind every entry, is `runs/timing_harvest.json`, produced by `campaign/harvest_timing.py` from the run records and logs.

What the members cost. Table S9.2 is the second campaign's cost record at all ten seeds, every member trained to the end of its schedule on one host. Thread counts differ across members because the host was shared with the rest of the campaign, so the comparable unit is core-minutes, wall clock times threads. At about 7900 core-minutes against 54 for the MSE MLP, the FNO costs roughly 150 times the cheapest neural member and returns 0.03 points of relative test error for it; the UNet costs about 115 times as much and is beaten by the MSE MLP and the refiner. That ratio is not a capacity effect: at 6.45M parameters the

Table S9.1: Compute record of the structural-mechanics members, as written by the campaign's own trainers and logs (all stages on CPU, six threads per task, five tasks sharing each instance). Network rows: the trainer's elapsed minutes from the first epoch to the final evaluation, as the range over seeds, with the scheduled epochs and the epoch of the kept checkpoint (range over the ten seeds). The FNO row gives seconds per epoch from the ten-epoch timer prints, minimum and median over all intervals (count in parentheses), since no seed completed the schedule. Kernel rows: the whole stage (fifteen grid cells and the refit) and the single factor-and-solve it repeats, as ranges, with the number of timed cells or folds in parentheses. The two columns are the two instance classes of the campaign, with the seed count behind each.

Stage	Epochs: scheduled / kept	Standard instances (6 seeds)	Larger instance (4 seeds)
Metric-loss MLP	400 / 39–79	129–158 min	26–35 min
MSE MLP	120 / 99–119	9.7–11.9 min	6.6–11.1 min
Refiner	100 / 99	9.9–13.1 min	7.9–33.7 min
FNO, per epoch	200 / 10–70	467 s min, 581 s median (88)	348 s min, 437 s median (5
Kernel ridge, grid and refit	—	17.0–26.0 min	6.1–7.4 min
one factor-and-solve, $n = 19000$	—	42–113 s (90)	20–35 s (60)
out-of-fold fold, $n = 14250$	—	32–66 s (24)	13–22 s (16)

Table S9.2: Cost record of the second campaign on a shared 40-core CPU host, every member trained to the end of its schedule; ranges over the ten seeds. Wall minutes are the trainer's own elapsed record; core-minutes is wall clock times the thread count (10 for the convolutional members, 10 to 16 for the MLPs, which is why those ranges are wide); the kernel stage does not write a thread count. Test errors are reflection-averaged where the member has a reflection, as in Table 4. The harvest is `campaign/collected/dgx/runtime_ledger.csv`, produced by `runtime_ledger.py` from the run records.

Member	Parameters	Epochs	Wall (min)	Core-min	Test error (%)
Kernel ridge, grid and refit	—	—	5.8–10.0	—	5.193–5.200
Metric-loss MLP	4.91M	400	8.8–19.2	132–192	4.809–4.858
MSE MLP	4.91M	120	2.6–5.8	39–58	4.703–4.719
Refiner	6.64M	100	2.6–6.1	41–61	4.721–4.742
FNO	6.45M	100	662–836	6618–8355	4.665–4.697
UNet	4.38M	100	573–689	5734–6892	4.683–4.793

FNO is 1.3 times the size of the MLP and smaller than the 6.64M-parameter refiner, and the UNet is the smallest member of the six at 4.38M. What separates them is the cost of a spectral or spatial convolution over a 41×41 field against that of a dense layer.

Stacking. Global weights: convex, initialized at the simplex minimizer of the measured second-moment matrix S (Proposition 6.1) and polished by a short random search on the validation metric. Per-pixel weights: affine in the members at each grid point, ridge parameter 10^{-3}, fitted on half the validation split and accepted only if they beat the global weights on the held-out half, then refitted on the full split.

Kernel stages. Matérn-5/2 on standardized inputs. The residual correction searches the scale grid $\{1, 2, 4\}\times$ the median pairwise distance (estimated on 2000 points) and the nugget grid $\{10^{-6}, 10^{-5}, 10^{-3}\}$ (scaled by n); the low-data chain of Table 3 uses $\{0.5, 1, 2, 4\}$ with nuggets $\{10^{-7}, 10^{-5}, 10^{-3}\}$. Both are tuned on validation using an 8000-sample subsample of the training set, refit at the chosen pair on the full training set in double precision. The pure kernel baseline (Table 2) uses the grid $\{0.5, 0.75, 1, 1.5, 2\}\times$median and nuggets $\{10^{-8}, 10^{-6}, 10^{-4}\}$ directly at $n = 19000$. Feature stage: penultimate activations of the ensemble members (FNO: spatially averaged pre-projection channels, 128; transformer: mean-pooled encoder tokens, 192; MLP: trunk output, 1024), concatenated, standardized, same kernel family and tuning.

Uncertainty. Posterior standard deviation (1) computed from the Cholesky factor of $K + n\lambda I$ by triangular solves. Conformal calibration uses a random 1000-sample subset of the test block that no training, checkpointing, stacking, or tuning step ever touched; the exact split-conformal quantile of the scores on that subset sets the band, and coverage is evaluated on the disjoint remainder. The hypotheses of Proposition S3.1 therefore hold as stated, with no approximate-exchangeability caveat.

S10 Numerical verification of the stated results

Every result stated in Section 6 is checked numerically, and the checks are released with the code. They come in two kinds. The first kind builds a synthetic object whose ground truth is known exactly and tests the stated inequality or identity on it: the certified bound of Theorem 6.9 is evaluated on a residual constructed inside the native space with exactly known norm, at several nuggets, where any pointwise violation beyond numerical precision would falsify the statement; the leave-half-out identity and unbiasedness of Proposition S1.2 are checked by direct enumeration; the effective-dimension identity of Lemma S5.1 is checked to machine precision and the closed form of Corollary S5.2 against the empirical spectrum of an i.i.d.-entry Gram matrix; and the rate mechanism of the heuristic calculation in Proposition S2.5 is exercised on a controlled warped target $G = g(A\,\cdot)$ with a known singular-value profile, where the adapted metric must dominate at every sample size and improve the empirical rate. All four pass: the largest relative violation of the optimal-recovery bound is -3.3×10^{-2}, that is, the bound holds with margin everywhere it was evaluated; the kernel-flow identity closes to 2.8×10^{-14}; the effective-dimension identity to 3.7×10^{-15} and its Marčenko–Pastur closed form to 4.4×10^{-5} across five nuggets spanning four orders of magnitude; and the adapted metric dominates the isotropic one at every sample size of the warped construction.

The second kind evaluates the statements whose subject is the pipeline itself on the pipeline's own released artifacts, at every seed. The symmetrization inequality of Proposition S1.1 is checked on every trained member. It is a statement about the expected loss, so a finite test sample may go the other way, and at ten seeds one does: over the thirty members that carry a reflection, averaging lowers test error in twenty-nine cases, by 0.16 points on the metric-trained network, 0.07 on the FNO and 0.02 on the normalized-MSE one, and raises it in the thirtieth by 0.001 points. That single reversal is the sampling noise the proposition permits and a single-seed check cannot see. The second-moment identity of Proposition 6.1 is measured on the validation residual matrix at each seed and its predicted risk compared with the realized risk of the weights it implies; the ratio is 0.938 ± 0.003 over ten seeds on five members and 0.938 ± 0.003 on six. The simplex minimization behind that comparison is solved by sequential quadratic programming under an explicit sum-to-one constraint, and the suite guards the solve with the cheapest check available: it evaluates the objective at the returned point, at every pure-member vertex and at a re-solved optimum, and requires the first to be no worse than the others. A clip-and-renormalize projected gradient, which is not a projection onto the simplex, fails that check on this matrix, stopping 0.8–3.6% above the minimum at near-vertex points. The bracket of Corollaries 6.3 and 6.4 is evaluated against the realized convex stack at each seed, with the one-sided bounds read off the measured S: on five members the normalized minimum $\min_w w^\top S w/\bar{e}^2$ is 0.980 ± 0.003 inside a bracket of $[0.949 \pm 0.005,\ 1.039 \pm 0.004]$, on six it is 0.972 ± 0.005 inside $[0.933 \pm 0.007,\ 1.032 \pm 0.004]$, and the inequalities hold at every seed. The signed optimum of Corollary 6.5 coincides with the convex one at every seed and for both member sets, the unconstrained minimizer having no negative coordinate. The bound of Theorem 6.6 is evaluated on the sixty predictors of Section 4.6 with the pool's smallest member error and smallest within- and between-architecture second moments: 4.814% against a hindsight convex optimum of 4.876% and a uniform mixture of 4.905%, root mean square, so the bound holds and sits six hundredths below the best mixture any rule could have chosen. The exchange rate of Proposition 6.7 is evaluated at the measured values of both campaigns in Supplement S6. The two-member rule of Proposition 6.1(i) is scored prospectively on every member pair of every seed rather than on selected examples — decided on the validation split and checked on the test block, 840 pairs over the thirty OCO-2 band-seeds. Section 5.2 reports what that scoring found, which is not a uniform verdict: the rule's accuracy is governed by the margin between the two quantities it compares. The selection conditions of Propositions S2.2, S2.4 and S1.5 are checked against the realized selections, with the measured constants b and W of Proposition S1.5 and Corollary S1.6 released next to the realized excesses of the affine stack and of the corrected surrogate. Both are capacity bounds: the affine bound evaluated at the measured b and W exceeds its realized excess by many orders of magnitude, and the corollary's right-hand side, whose remaining constant is the analytic $\kappa_{\mathrm{an}} = 500$ of Lemma S1.7 (half the smallest nugget on the grid to the power $-1/2$), is larger still; the constants are released for what they show about the bounds rather than as a guarantee. The realized scale of the correction weights is measured separately: the largest ℓ^1 norm of $c_\gamma(u)$ over the grid and over a fixed 500-point test probe is 32.2 ± 2.9 over the ten seeds, between 28.9 and 37.3, falling to 3.1 at the largest nugget and length scale of the grid, and recomputing seed 0 on a second machine reproduces its value to 10^{-12}. Being a supremum over a finite probe, that value is a lower bound on $\sup_{u,\gamma} \|c_\gamma(u)\|_1$ and not a constant the corollary can be evaluated at, and the corollary is evaluated at κ_{an}. The probe value sits a factor of about fifteen below κ_{an}, and every per-configuration probe value respects the lemma's bound $\frac{1}{2}\lambda^{-1/2}$ at its own nugget, the closest approach being 0.27 of the bound, at $\lambda = 10^{-3}$. The coverage of Proposition S3.1 is computed at every seed and each observed value placed against the exact Beta law of Remark S3.2: at the 90% level ten of ten fall inside the central band $[88.1\%, 91.8\%]$, and at the 95% level ten of ten inside $[93.6\%, 96.3\%]$; on the six-member pipeline the counts are ten of ten and nine of ten, the tenth sitting on the upper edge of the 95% band. Finally the data-scaling sweep of the two raw-input kernel rows is refit at nested training sizes and at ten seeds per band. The approximate rank of the sensitivity map is 12 of 20 on O2 and 13–14 of 24 on the two CO_2 bands, a rank ratio between 0.54 and 0.60, so the heuristic calculation of Proposition S2.5 would permit a rate gain here and not merely a constant; the sweep finds the constant and not the rate, and Section 5.1 reports it that way.

The floor theorem, the exchange rate, the sharpened weight bound, the minimax identity and the signed-stack corollary have their own released check, fifteen tests in all, in `campaign/verify_floor_and_certificate.py`. The block floor of Theorem 6.6 is an identity in the exact block-correlated model, and the identity closes to 3×10^{-18} over two thousand random convex weight vectors; on a perturbed pool, and on the released sixty-predictor matrix, twenty thousand random convex mixtures never pass the bound evaluated with the pool's minima, and the mean-entry form of the display returns the uniform mixture's value exactly. The two partial derivatives of Proposition 6.7 agree with central differences to 4×10^{-9} over two thousand interior points, both are positive there, and along the stated direction the change in the optimal risk is second order in the step. The sharpened bound of Lemma S1.7 holds on fifteen hundred random Matérn designs and evaluation points, the largest ratio to the bound being 0.87, and the cosine-kernel construction attains it to 7×10^{-15}. The identity of Theorem 6.10 closes to 4×10^{-14}, the ordering $P_0 \leq \widetilde{P}_\lambda \leq P_\lambda$ holds on every draw, the residual that the proof of Theorem 6.9 exhibits attains the certified bound with equality to 10^{-14}, and the signed optimum of Corollary 6.5 matches its closed form and never lies above a sampled convex mixture.

The bound comparisons are reported as measured constants, bound value and realized excess side by side; where an oracle risk is needed it is estimated by a deterministic refit on the test arrays, which can only make the reported ratios conservative. The synthetic checks report pass or fail against explicit tolerances; the artifact checks report the measured quantities themselves, together with the number of seeds behind each.

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